

Cell-Cycle-Targeting MicroRNAs as Therapeutic Tools against Refractory Cancers

Highlights

- Characterization of human microRNAs which target the cell-cycle machinery
- Profiling cell-cycle-targeting miRNAs against 122 human cancer cell lines from CCLE
- Algorithm to predict the response of tumors to cell-cycle-targeting miRNAs
- In vivo delivery of cell-cycle-targeting miRNAs inhibits cancer growth

Authors

Per Hydbring, Yinan Wang, Anne Fassl, ..., Daniel G. Anderson, Cheng Li, Piotr Sicinski

Correspondence

cheng_li@pku.edu.cn (C.L.),
peter_sicinski@dfci.harvard.edu (P.S.)

In Brief

By performing screens for miRNAs targeting cell-cycle proteins, Hydbring et al. identify a class of miRNAs that target multiple cyclins and CDKs. Nanoparticle delivery of these miRNAs inhibits tumor growth in several xenograft models, including treatment-refractory patient-derived xenografts.



Cell-Cycle-Targeting MicroRNAs as Therapeutic Tools against Refractory Cancers

Per Hydbring,^{1,2,3} Yinan Wang,⁴ Anne Fassl,^{1,2} Xiaoting Li,⁵ Veronica Matia,^{1,2} Tobias Otto,^{1,2} Yoon Jong Choi,^{1,2} Katharine E. Sweeney,^{1,2} Jan M. Suski,^{1,2} Hao Yin,⁶ Roman L. Bogorad,⁶ Shom Goel,^{1,7} Haluk Yuzugullu,^{1,8} Kevin J. Kauffman,⁶ Junghoon Yang,⁶ Chong Jin,⁴ Yingxiang Li,⁹ Davide Floris,^{1,2} Richard Swanson,¹⁰ Kimmie Ng,⁷ Ewa Sicinska,¹¹ Lars Anders,¹ Jean J. Zhao,^{1,8} Kornelia Polyak,^{7,12} Daniel G. Anderson,^{6,13,14,15} Cheng Li,^{4,*} and Piotr Sicinski^{1,2,16,*}

¹Department of Cancer Biology, Dana-Farber Cancer Institute

²Department of Genetics, Harvard Medical School

Boston, MA 02215, USA

³Department of Oncology-Pathology, Karolinska Institutet, 17176 Stockholm, Sweden

⁴Peking-Tsinghua Center for Life Sciences, Academy for Advanced Interdisciplinary Studies, School of Life Sciences, Center for Life Sciences and Center for Statistical Science, Peking University, Beijing 100871, China

⁵School of Life Sciences, Tsinghua University, Beijing 100084, China

⁶David H. Koch Institute for Integrative Cancer Research, Massachusetts Institute of Technology, Cambridge, MA 02142, USA

⁷Department of Medical Oncology, Dana-Farber Cancer Institute, Boston, MA 02215, USA

⁸Department of Biological Chemistry and Molecular Pharmacology, Harvard Medical School, Boston, MA 02115, USA

⁹Department of Bioinformatics, School of Life Science and Technology, Tongji University, Shanghai 200092, China

¹⁰Department of Surgery, Brigham and Women's Hospital, Boston, MA 02115, USA

¹¹Department of Oncologic Pathology, Dana-Farber Cancer Institute

¹²Department of Medicine, Harvard Medical School

Boston, MA 02215, USA

¹³Department of Chemical Engineering

¹⁴Institute for Medical Engineering and Science

¹⁵Harvard-MIT Division of Health Sciences & Technology

Massachusetts Institute of Technology, Cambridge, MA 02142, USA

¹⁶Lead Contact

*Correspondence: cheng_li@pku.edu.cn (C.L.), peter_sicinski@dfci.harvard.edu (P.S.)

<http://dx.doi.org/10.1016/j.ccell.2017.03.004>

SUMMARY

Cyclins and cyclin-dependent kinases (CDKs) are hyperactivated in numerous human tumors. To identify means of interfering with cyclins/CDKs, we performed nine genome-wide screens for human microRNAs (miRNAs) directly regulating cell-cycle proteins. We uncovered a distinct class of miRNAs that target nearly all cyclins/CDKs, which are very effective in inhibiting cancer cell proliferation. By profiling the response of over 120 human cancer cell lines, we derived an expression-based algorithm that can predict the response of tumors to cell-cycle-targeting miRNAs. Using systemic administration of nanoparticle-formulated miRNAs, we inhibited tumor progression in seven mouse xenograft models, including three treatment-refractory patient-derived tumors, without affecting normal tissues. Our results highlight the utility of using cell-cycle-targeting miRNAs for treatment of refractory cancer types.

Significance

Targeting cell-cycle machinery represents an attractive anti-cancer therapeutic strategy, and chemical inhibitors of cyclin-CDK kinases are in clinical trials. Here, we identified a class of miRNAs targeting multiple cyclin/CDKs and propose that these miRNAs might be superior to currently available therapeutic compounds. We found that cell-cycle-targeting miRNAs are very potent against triple-negative breast cancer, one of the most aggressive breast cancer types. We provide an approach to select miRNAs that are particularly efficacious against a given tumor, thereby allowing individualized therapies. We demonstrate that cell-cycle-targeting miRNAs can be administered to tumor-bearing animals, where they inhibit growth of patient-derived tumors that are resistant to available therapies. Hence, this study suggests a strategy of targeting aggressive human tumors.

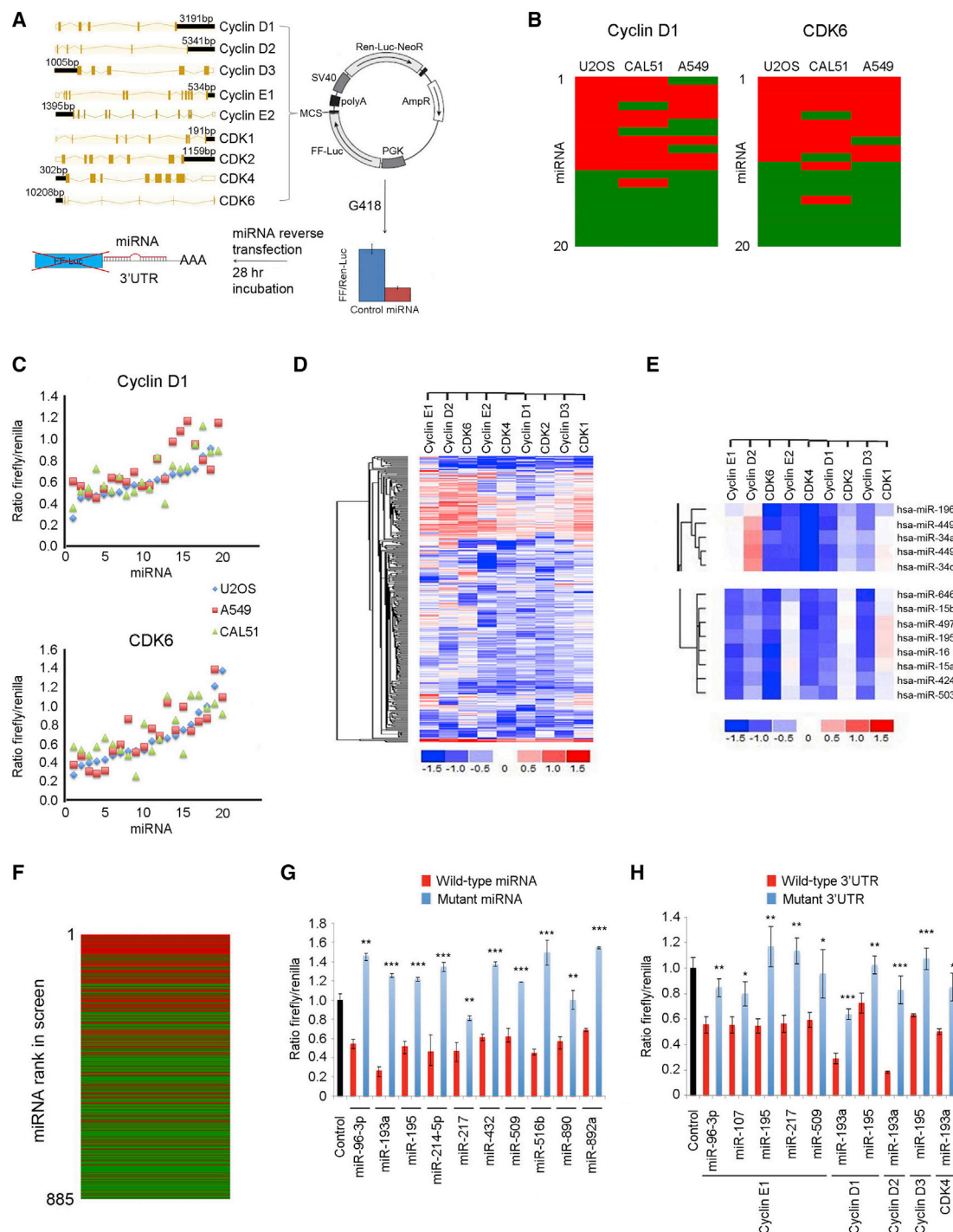


Figure 1. Genome-Wide Screens for miRNAs Targeting Cyclins and CDKs

(A) Screening approach. Gene depictions are adapted from Ensembl, release 87. Black portions of genes denote cloned 3'UTR regions.

(B) Comparison of the ability of 20 selected miRNAs to target cyclin D1 or CDK6 3'UTRs in assays performed in U2OS, CAL51, and A549 cells. Each horizontal row depicts a different miRNA, which were arranged in the same order as in (C). Red denotes targeting ($\geq 40\%$ repression of firefly to renilla luciferase ratio); green, no targeting. For each miRNA and cell line, $n = 3$.

(C) Comparison of firefly to renilla luciferase ratios for 20 selected miRNAs in assays performed in U2OS, CAL51, and A549 cells expressing cyclin D1 or CDK6 3'UTRs (as in B). miRNAs were arranged according to increasing ratios in the original U2OS-based screen. For each miRNA and cell line, $n = 3$.

(D) Hierarchical clustering of the results of our nine screens. Each vertical column corresponds to a different screen, each horizontal row depicts a different miRNA. For each miRNA, $n = 3$.

(legend continued on next page)

INTRODUCTION

The proliferation of mammalian cells is driven by cyclins and their catalytic partners, cyclin-dependent kinases (CDKs). Cyclin-CDK complexes phosphorylate cellular proteins, thereby driving cell-cycle progression. Stimulation of cells with growth factors induces the expression of D-type cyclins (cyclins D1, D2, and D3), which bind and activate CDK4 or CDK6. Later during the G1 phase, E-type cyclins (cyclins E1 and E2) become upregulated and activate CDK2 (and to a lesser extent CDK1). Cyclin E-CDK2 complexes regulate entry of cells into the DNA synthesis (S phase). Further progression of cells through the S phase is driven by cyclin A2, which partners with CDK2. Later during mitotic prophase, cyclin B translocates to the nucleus and activates CDK1. Cyclin B-CDK1 kinase drives mitotic events such as spindle pole assembly, chromosome condensation, and nuclear envelope breakdown (Malumbres and Barbacid, 2009).

A comprehensive analysis of human cancers revealed that genes encoding cyclins and CDKs belong to the most frequently amplified loci (Beroukhi et al., 2010). The importance of overexpression of D-type cyclins in pathogenesis of human cancer has been particularly well established (Musgrove et al., 2011). Analyses of mouse genetic models revealed the requirement for specific cyclins and CDKs in development and maintenance of a wide array of tumor types (Malumbres and Barbacid, 2009).

Collectively, these observations firmly established essential roles for cyclins and CDKs in tumorigenesis and led to the development of several CDK inhibitors. Some of these compounds are currently in clinical trials, and three CDK4/6 inhibitors (palbociclib, abemaciclib, and ribociclib) received a “Breakthrough Therapy” designation from the US Food and Drug Administration and were approved for treatment of estrogen-receptor-positive breast cancers (Asghar et al., 2015; Finn et al., 2015). However, the clinical success of targeting the cell-cycle machinery has been limited so far. For instance, palbociclib was shown to significantly prolong progression-free survival of cancer patients, but it had no major impact on overall survival (Finn et al., 2015). A likely reason for this disappointing outcome is that redundant cyclins and CDKs compensate for the inhibition of CDK4 and CDK6, thereby allowing tumor progression. For this reason, agents that target multiple cyclins and CDKs might offer a therapeutic advantage by preventing compensatory upregulation of cell-cycle kinases.

MicroRNAs (miRNAs) have been recognized for their potential in cancer therapeutics, and multiple miRNAs were suggested to either play tumor-suppressive or tumor-promoting roles (Adams

et al., 2014; Hayes et al., 2014). miRNAs bind their target transcripts via the 3' UTRs of mRNAs. miRNAs do not extinguish expression of their targets but reduce their levels, with individual miRNAs targeting many different transcripts (Bartel, 2009). We hypothesized that such a broad dampening of expression of several cell-cycle proteins might allow to selectively block proliferation of cancer cells without having major effects on their non-transformed counterparts. Moreover, targeting several cell-cycle proteins at once could be beneficial in cancer therapy since it would give less room for cell-cycle compensatory mechanisms that may lead to acquired resistance to CDK inhibition.

Although numerous studies postulated targeting cyclins and CDKs by miRNAs in cancer treatment (Bonci et al., 2008; Johnson et al., 2007; Kota et al., 2009), no attempt was made to systematically delineate miRNAs regulating these proteins. In this study, we performed nine genome-wide screens to identify the full range of miRNAs regulating major cell-cycle proteins.

RESULTS

Genome-Wide Screens and Their Validation

In order to determine the full repertoire of miRNAs directly regulating cell-cycle cyclins and CDKs, we cloned 3'UTRs of cyclins D1, D2, D3, E1, E2, as well as CDK1, CDK2, CDK4, and CDK6 downstream of the firefly luciferase gene into a dual luciferase reporter vector. The vector also encoded renilla luciferase driven by the SV40 promoter (Figure 1A). Vectors containing 3'UTRs of cyclins or CDKs were subsequently stably expressed in U2OS cells, thereby generating nine reporter cell lines. Expression of miRNAs targeting the 3'UTR of a given cyclin or CDK in these cell lines is expected to repress the expression of the firefly luciferase while leaving renilla luciferase levels intact, and hence to decrease the firefly to renilla ratio.

Reporter cell lines stably expressing each of nine different 3'UTR luciferase constructs were plated in 96-well plates and transfected with human miRNA mimic library containing 885 annotated human miRNAs, one miRNA per well.

The firefly to renilla luciferase ratios were determined 28 hr post transfection (Figure 1A and Table S1). All screens were performed in three technical replicates with all replicates displaying correlations of 0.9 or higher (Figure S1A).

Since in our screens all miRNA mimics as well as the reporter system were of the exogenous origin, we considered it unlikely that the choice of the cell line (U2OS cells) would affect the results of the screens. Nevertheless, to exclude this possibility,

(E) A zoom-in from (D) to illustrate that miRNAs with identical seed sequences cluster together in their targeting pattern of cyclin/CDK 3'UTRs. Upper panel, the miR-34/449-family; lower panel, the extended miR-15/16-family. For each miRNA, $n = 3$.

(F) Overlay of the results of our cyclin D1 3'UTR screen with TargetScan predictions. Each horizontal row corresponds to a different miRNA. miRNAs were arranged based on the results of our screen, with miRNA most potently repressing cyclin D1 3'UTR at the top. miRNAs predicted by TargetScan to target cyclin D1 3'UTR are marked in red, miRNAs predicted not to target cyclin D1 3'UTR are in green. For each miRNA, $n = 3$.

(G) Firefly to renilla luciferase ratios in U2OS cells expressing cyclin D1 3'UTR-based reporter, upon transfection of the indicated wild-type miRNAs (red bars), or the corresponding miRNAs containing point mutations within their seed sequences (blue bars). Black bar, control (scrambled) miRNA. For each miRNA, $n = 3$. (H) U2OS cells were engineered to express mutant versions of 3'UTRs for cyclin E1, D1, D2, D3, or CDK4, containing mutations or deletions within miRNA target sequences. Shown are firefly to renilla luciferase ratios upon expression of the indicated miRNAs in cells containing wild-type (red bars) or mutant 3'UTRs (blue bars). Black bar, control (scrambled) miRNA. For each miRNA, $n = 3$.

Data are mean values $\pm$ SD. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ using unpaired t test.

See also Figures S1–S3, and Table S1.



(B) Enrichment analysis showing the number of miRNAs targeting multiple cyclins/CDKs observed in our screens (red) and expected results based on permutation analysis (blue).

(legend continued on next page)

we first determined that there was no correlation between the results of our screens and the levels of endogenous cyclins, CDKs, or miRNAs in U2OS cells (Figures S1B–S1G). Moreover, we validated our screening results using two additional epithelial cell lines, CAL51 (breast cancer) and A549 (lung adenocarcinoma); please see Figures S2A–S2E for cell-cycle characterization of these cells. As we did for U2OS cells, we engineered CAL51 and A549 cells to stably express the luciferase reporters linked to 3'UTRs of nine cyclins or CDKs. We then compared the response of U2OS-, CAL51-, and A549-based reporter cell lines to 20 selected miRNAs. We found that the results were highly reproducible between the three cell lines. Thus, the overall reproducibility between screening results in U2OS and CAL51 cells was 79.4%, between U2OS and A549 was 78.3%, and between A549 and CAL51 was 75.6% (Figures 1B, 1C, S3A, and S3B). We concluded that the choice of U2OS cells for the screens did not significantly influence the results.

We performed unsupervised clustering of the results of all nine screens (Figures 1D and 1E). Strikingly, we found that miRNAs containing a common seed sequence invariably clustered together, i.e., they produced almost identical results across the nine screens, as seen for instance for the miR-34/449 family (Figure 1E, upper panel), and the extended miR-15/16 family (Figure 1E, lower panel). This observation provided an additional confirmation for the reliability of our screens.

We next compared the results of our nine screens with *in silico* predicted targeting using five softwares: TargetScan, miRanda, miRDB, miRWalk, and TargetSpy. Each of these algorithms predicts targeting of transcripts by miRNAs, based on the 3'UTR sequences of the genes (Figure S3C). We found TargetScan to be most accurate in its ability to predict targeting, followed by miRanda (Figures S3D–S3F). We observed a strong correlation between TargetScan context score (which predicts the likelihood of targeting) and the results of our screens (Figures S3G and S3H). Despite overall correlation, several miRNAs that were predicted to target a given cyclin or CDK did not score in our screens (Figure 1F), highlighting the limitation of the TargetScan predictive value.

A selected number of “hits” from our screens were further validated using mutant miRNAs containing point mutations within their seed sequences (Figure 1G), as well as mutant 3'UTRs with mutations or deletions within miRNA target sequences (Figure 1H). In all cases examined, these mutations strongly diminished the ability of miRNAs to repress luciferase expression (Figures 1G and 1H). Lastly, we confirmed the ability of selected miRNAs to reduce the expression of endogenous cyclin and CDK transcripts in 12 cancer cell lines (see below). We concluded that miRNAs identified in our screens indeed repress expression of cyclins and CDKs.

Identification of Cell-Cycle-Targeting miRNAs

We constructed a heat map of all miRNAs targeting at least one cyclin or CDK. These analyses revealed the presence of a group

of miRNAs that target all or nearly all cell-cycle proteins analyzed (Figure 2A). Importantly, the number of these cell-cycle-targeting miRNAs was significantly higher than the number predicted from random permutations (Figure 2B). Thus, we found 16 miRNAs targeting 5 cyclins or CDKs, compared with 0.26 predicted from random permutations, 6 miRNAs targeting 6 cyclins/CDKs (predicted 0.01), 4 miRNAs targeting 7 cyclins/CDKs (predicted 0.0007), 3 miRNAs targeting 8 cyclins/CDKs (predicted 0), and 1 miRNA targeting 9 cyclins/CDKs (predicted 0) (Figure 2B); p values $<10^{-4}$ for all cases. In total, we enumerated 30 miRNAs targeting at least 5 cyclins/CDKs and 14 miRNAs targeting at least 6 cyclins/CDKs (Figures 2A and 2B; Table S1). In contrast, the number of miRNAs uniquely targeting individual cyclins or CDKs was not higher than predicted from random permutations (Figure S3I).

Also pairwise comparisons of all screening data revealed that miRNAs targeting one cell-cycle protein were significantly enriched for targeting other cell-cycle proteins (Figure 2C). Collectively, these analyses revealed that the mammalian genome contains a previously unanticipated class of miRNAs that target multiple components of the core cell-cycle machinery. We called these miRNAs cell-cycle-targeting miRNAs.

In order to provide a quantitative measure of the ability of miRNAs to repress the cell-cycle machinery, for each miRNA we calculated the average repression value across the nine screens. Using this criterion, 16 miRNAs repressed all nine 3'UTRs by an average of 40% or more, while 60 miRNAs showed at least average 30% repression across the nine screens (Table S1).

We next asked whether cell-cycle-targeting miRNAs might also target the other three major cyclins that were not included into our screens, namely cyclins A2, B1, and B2. To address this point, we generated U2OS reporter cell lines stably expressing luciferase constructs containing 3'UTRs of cyclins A2, B1, or B2, and tested their response to ten randomly selected cell-cycle-targeting miRNAs. We found that several cell-cycle-targeting miRNAs repressed the expression of cyclin A2, B1, and B2 reporter constructs. In contrast, ten randomly selected control, non-cell-cycle-targeting miRNAs had essentially no effect (Figures 2D–2F). We also determined that cell-cycle-targeting miRNAs were not enriched in targeting other pro-proliferative pathways, beside cyclins and CDKs (Table S2). We concluded that the mammalian genome expresses a distinct class of miRNAs that can regulate expression of up to 12 major cyclins and CDKs.

Computational Analyses of Cell-Cycle-Targeting miRNAs

Using data from The Cancer Genome Atlas (TCGA), we analyzed the expression of cell-cycle-targeting miRNAs across 4,807 human tumor samples representing 18 tumor types. We searched for correlation/anti-correlation between the expression of miRNAs and the levels of transcripts encoding cyclins and CDKs. We found that miR-195-5p showed the strongest anti-correlation with expression of transcripts encoding cyclins

(D–F) Results of cyclin A2 (D), B1 (E), and B2 (F) 3'UTR luciferase-reporter assays. Shown are comparisons between the effect of ten cell-cycle-targeting miRNAs and 10 miRNAs that do not target cyclins/CDKs (chosen based on the results of our screens). Vertical axis depicts the ratios of firefly/renilla luciferase. Values observed in cells transfected with control miRNA were set at 1.0. miRNAs repressing firefly luciferase by at least 40% are marked in red. Data are mean values $\pm$ SD. For each miRNA, $n = 3$.

See also Table S2.

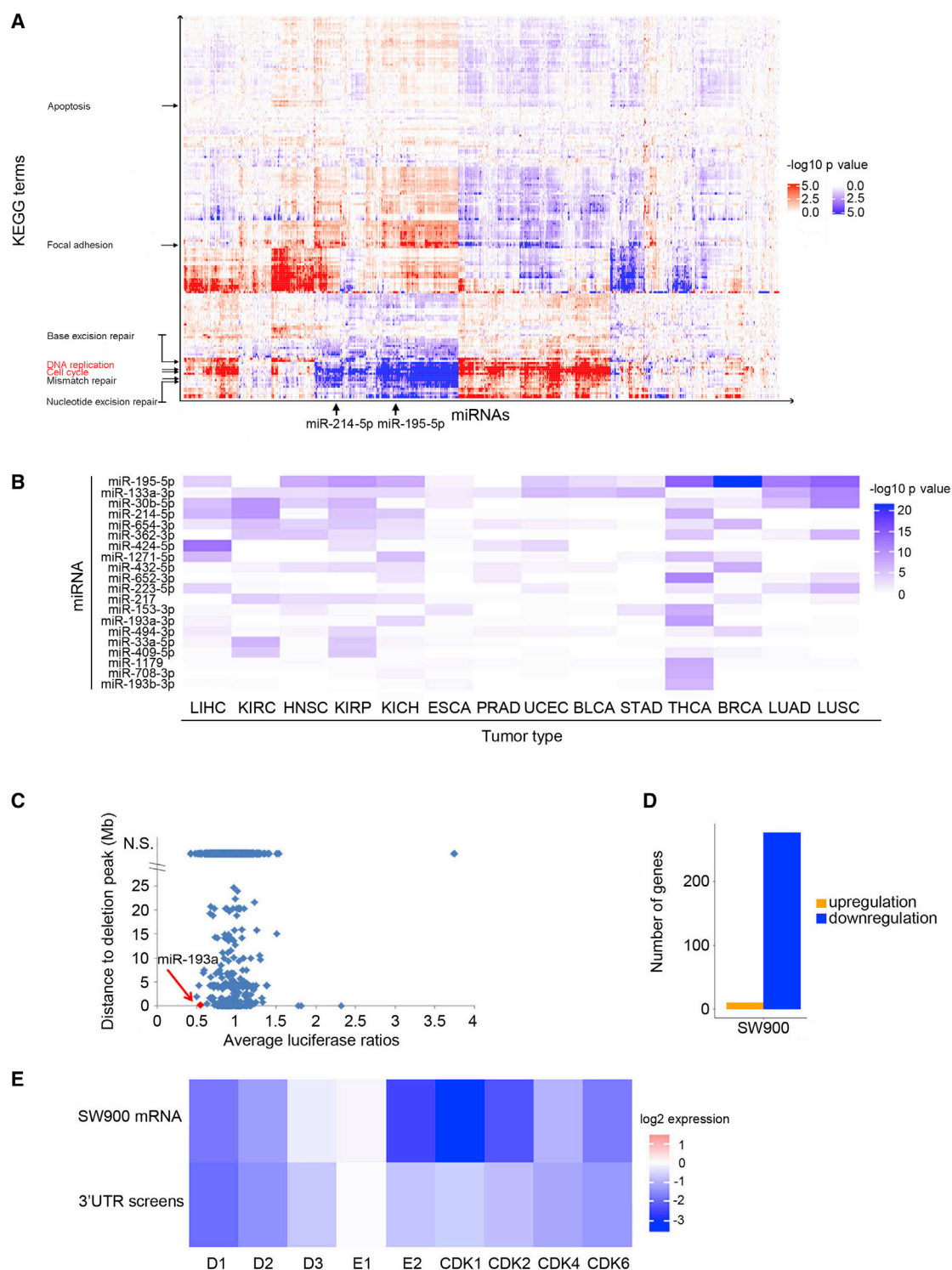


Figure 3. Computational Analyses of Cell-Cycle-Targeting miRNAs

(A) Guilt-by-association analysis of all miRNAs. Each horizontal row corresponds to a different KEGG pathway, and each vertical column denotes a different miRNA. The heatmap shows $-\log_{10} p$ values for the enrichment of KEGG pathways among transcripts correlated with a given miRNA across 4,807 human tumor samples representing 18 tumor types, with colors representing positive correlation (red) and anti-correlation (blue). p values were calculated using a hypergeometric test.

(B) A heatmap of comparisons of the indicated miRNAs levels between primary tumors versus adjacent, normal tissue from the same patient (data from TCGA). The top 20 cell-cycle-targeting miRNAs most downregulated in tumor samples are shown. Color scale depicts $-\log_{10} p$ value, with most significant

(legend continued on next page)

E1, E2, CDK1, CDK2, and CDK4, and also displayed a strong anti-correlation with expression of cyclins D1, D2, D3, and CDK6 (Table S3), suggesting that it might play a role in regulating the levels of these proteins in cancer cells.

We also searched for correlation/anti-correlation between expression of all miRNAs and expression of all protein-coding transcripts in 4,807 human tumor samples. The correlating transcripts were then analyzed for enrichment in KEGG pathway categories. This guilt-by-association analysis revealed that expression of some cell-cycle-targeting miRNAs, such as miR-195-5p and miR-214-5p, strongly anti-correlated with expression of genes belonging to cell-cycle and DNA replication pathways (Figure 3A).

We hypothesized that cell-cycle-targeting miRNAs, through their virtue of repressing the cell-cycle machinery, might display growth-suppressive properties. Consequently, we predicted that expression of some of these miRNAs might be silenced during tumorigenesis. To test this prediction, we used the TCGA database to compare the expression of all cell-cycle-targeting miRNAs in 14 types of human tumors versus adjacent healthy tissue from the same patient. We found that of all miRNAs, miR-195-5p was the one most significantly downregulated in 11 of 14 tumor types (Figure 3B; p values from 1.77×10^{-12} to 0.011 for significant downregulation, Wilcoxon test).

In addition, we intersected the results of our nine screens with the TCGA registry of deleted regions across many human tumor types and searched for cell-cycle-targeting miRNAs that are commonly deleted in human cancers (Figure 3C). These analyses revealed that the miR-193a gene, which encodes one of the most potent cell-cycle-regulating miRNAs, miR-193a-3p, is frequently deleted in several types of human tumors. We also identified numerous cell lines displaying copy number deletions of miR-193a (Figure S4A). Among them, a lung squamous cell carcinoma cell line, SW900, harbors a very narrow, focal homozygous deletion encompassing the miR-193a gene (Figures S4A and S4B).

To test the response of SW900 cells to re-introduction of the deleted miRNA, we transfected double-stranded mimic of miR-193a into SW900 cells and analyzed transcript abundance by RNA sequencing. Consistent with the well-established repressive effect of miRNAs on gene expression, nearly all transcripts that were affected by re-introduction of miR-193a displayed reduced levels (Figure 3D). Strikingly, the repressed targets were strongly enriched in the cell-cycle category and included cyclins and CDKs (Figures 3E and S4C). Consistent with inhibition of the cell-cycle machinery, expression of miR-193a blocked proliferation of SW900 cells (data not shown).

Cell-Cycle-Targeting miRNAs Inhibit Proliferation of Cancer Cells

We asked whether other cell-cycle-targeting miRNAs might also have an anti-proliferative effect on cancer cells. To test this, we transiently expressed a panel of 11 miRNAs targeting 6 to 9 cyclins/CDKs (Figure 4A) in human osteosarcoma U2OS cells and monitored cell number expansion over 6 days. We found that all 11 cell-cycle-targeting miRNAs strongly inhibited proliferation of U2OS cells (Figures 4B and 4C). Importantly, point mutations within miRNA seed sequences abolished the ability of miRNAs to inhibit cell growth, confirming the specificity of the effect (Figure 4D).

In parallel, we tested 8 miRNAs that target only 1–3 cyclins/CDKs, and 13 miRNAs that do not target any of the analyzed cell-cycle proteins (Figure 4A). miRNAs targeting 1–3 cyclins/CDKs had only a mild effect on cell growth, while non-cyclin/CDK-targeting miRNAs had no overall effect (Figures 4B and 4C). We concluded that cell-cycle-targeting miRNAs display strong anti-proliferative properties when ectopically expressed in cancer cells.

Profiling the Response of Human Cancer Cell Lines from CCLE to Cell-Cycle-Targeting miRNAs

In order to further investigate the anti-proliferative properties of cell-cycle-targeting miRNAs, we profiled the response of 122 human cancer cell lines from the Cancer Cell Line Encyclopedia (CCLE), representing 12 separate cancer types, to four selected miRNAs. For profiling, we chose miR-195-5p, miR-193a-3p, and miR-214-5p, which were highlighted above, as well as miR-890, the latter being chosen based on its identification as one of the top hits in our screens (Table S1).

We transiently expressed each of the four miRNAs in 122 cell lines and monitored for reduction in cell number expansion (Figure 5A), induction of apoptosis (Figure 5B), and induction of cellular senescence (Figures S5A). We found that all four miRNAs had a profound effect on cancer cell proliferation (Figures 5A, 5C, and S5B; Table S4). In several cell lines expression of cell-cycle-targeting miRNAs also triggered cancer cell death (Figure 5B and Table S4). None of the cell-cycle-targeting miRNAs were found to induce senescence in a significant fraction of cell lines (Figure S5A and Table S4).

Among all cancer types analyzed, cell lines corresponding to gastric and triple-negative breast cancers (TNBCs) displayed the strongest response to expression of cell-cycle-targeting miRNAs, with miR-193a-3p being the most effective in suppressing cell numbers across all tumor types (Figures 5C and S5B). TNBCs underwent a significant, often over 10-fold

downregulation in deep blue. LIHC, liver hepatocellular carcinoma; KIRC, kidney renal clear cell carcinoma; HNSC, head and neck squamous cell carcinoma; KIRP, kidney renal papillary cell carcinoma; KICH, chromophobe renal cell carcinoma; ESCA, esophageal carcinoma; PRAD, prostate adenocarcinoma; UCEC, uterine corpus endometrial carcinoma; BLCA, bladder urothelial carcinoma; STAD, stomach adenocarcinoma; THCA, thyroid carcinoma; BRCA, breast invasive carcinoma; LUAD, lung adenocarcinoma; LUSC, lung squamous cell carcinoma. p values were calculated using the Wilcoxon test.

(C) Identification of cell-cycle-targeting miRNAs that are deleted in human cancers. The horizontal axis denotes average firefly/renilla luciferase ratios from the nine screens, with lower values (on the left) representing stronger repression. The vertical axis shows the distance between the locus encoding a given miRNA and the nearest deletion peak (from TCGA).

(D) The number of genes with altered expression levels following re-introduction of miR-193a-3p into SW900 cells.

(E) Comparison of repression of endogenous cyclins/CDKs transcript levels observed upon re-introduction of miR-193a-3p into SW900 cells (upper row) versus the effect of miR-193a-3p in nine 3'UTR luciferase screens (lower row). As expected from the results of our screens, expression of miR-193a-3p repressed the levels of all cyclins and CDKs except for cyclin E1. D1, cyclin D1; D2, cyclin D2; D3, cyclin D3; E1, cyclin E1; E2, cyclin E2.

See also Figure S4 and Table S3.

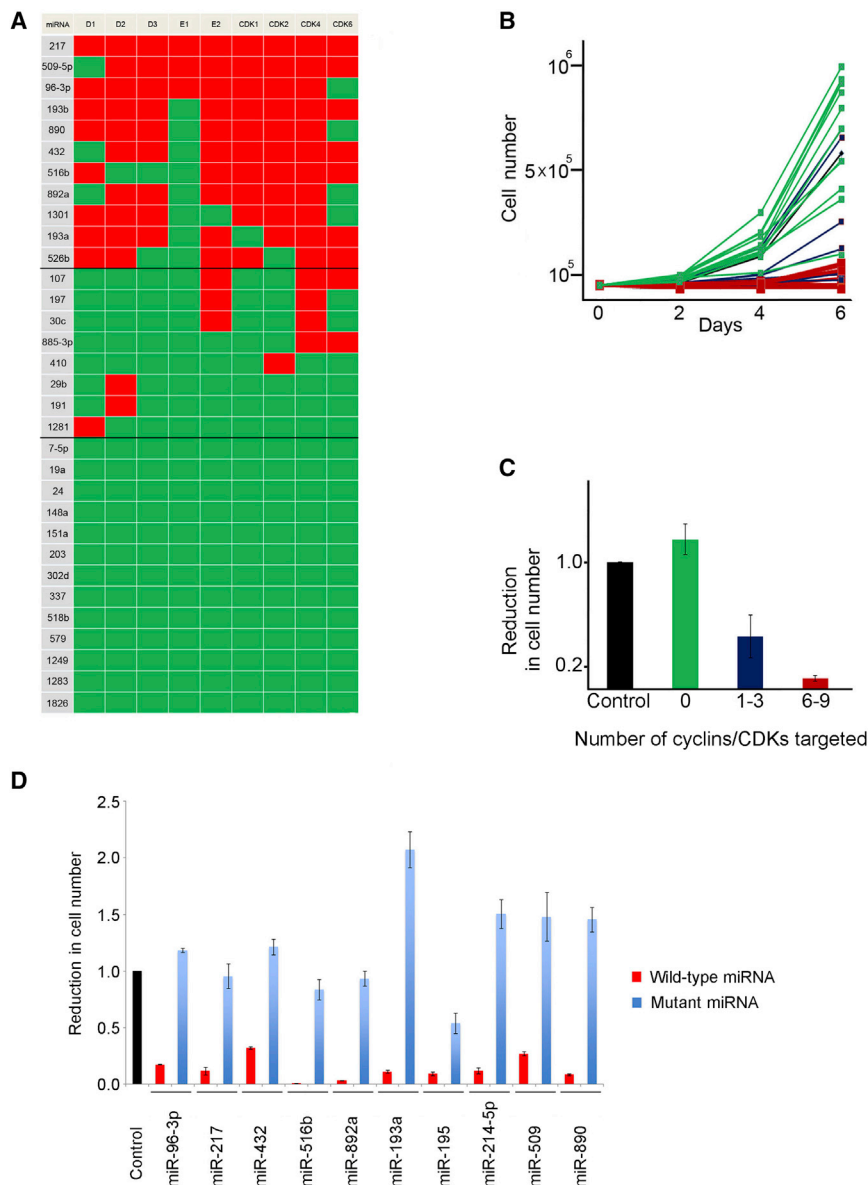


Figure 4. Cell-Cycle-Targeting miRNAs Block Cancer Cell Expansion

(A) A heatmap showing a panel of 11 selected miRNAs targeting 6–9 cyclins or CDKs, 8 miRNAs targeting 1–3 cyclins/CDKs, and 13 miRNAs not targeting any cyclins/CDKs, which were used for assays shown in (B). Each horizontal row corresponds to the indicated miRNA, each vertical column depicts the results of the indicated 3'UTR screen. D1, cyclin D1; D2, cyclin D2; D3, cyclin D3; E1, cyclin E1; E2, cyclin E2. Red indicates targeting ($\geq 40\%$ reduction of firefly/renilla ratio); green, no targeting. For each miRNA, $n = 3$.

(B) U2OS cell number expansion over 6 days following ectopic expression of miRNAs from (A). Red lines denote cell numbers in cultures transfected with miRNAs targeting 6–9 cyclins/CDKs; blue lines, miRNAs targeting 1–3 cyclins/CDKs; green lines, miRNAs not targeting cyclins/CDKs. Black line depicts cell counts for U2OS cells transfected with miRNA negative control. For each miRNA, $n = 3$.

(C) Reduction in cell number expansion in cells transfected with miRNAs targeting no cyclins/CDKs (green bar, 0), 1–3 cyclins/CDKs (blue bar, 1–3), and 6–9 cyclins/CDKs (red bar, 6–9). The reduction in cell number was defined as the ratio of cell numbers observed in cells transfected with this miRNA to cell numbers observed in cells transfected with control miRNA. Data are mean values $\pm$ SD. For each miRNA, $n = 3$.

(D) Reduction in cell number expansion in U2OS cells transfected with the indicated wild-type miRNAs (red bars), or the corresponding miRNAs containing point mutations within their seed sequences (blue bars). The reduction in cell number was defined as the ratio of cell numbers observed in cells transfected with this miRNA to cell numbers observed in cells transfected with control miRNA. Data are mean values $\pm$ SD. For each miRNA, $n = 3$.

reduction in cell number expansion, with several TNBC cell lines also displaying a pronounced apoptotic response (Figures 5A and 5B; Table S4); see Figure S5C for expression of cell-cycle proteins in TNBC. Importantly, we verified that ectopic expression of miR-193a-3p in TNBC led to downregulation of predicted cell-cycle targets (Figures S5D and S5E).

To test whether the observed effects on cancer cell expansion and apoptosis were mediated via the ability of cell-cycle-targeting miRNAs to repress the expression of target cyclins/CDKs, we engineered four TNBC cell lines to ectopically express cDNAs encoding cyclins D1, D3, and E2 (Figure S5E), which represent targets of miR-193a-3p (Table S1). Importantly, the three cyclin cDNAs lack 3'UTRs, and hence they are resistant to targeting by miRNAs. In contrast to parental cells, where miR-193a-3p blocked cancer cell growth, cells expressing miRNA-resistant cyclins were refractory to the anti-proliferative effect of miR-193a-3p, or, in the case of MDA-MB-468 cells, the anti-proliferative effect was

blunted (Figure 5D). Moreover, expression of miRNA-resistant cyclins in four breast cancer cell lines largely prevented induction of apoptosis by miR-193a-3p in vitro (Figure S5F), and prevented apoptosis of tumor cells in vivo (see below). These observations indicate that targeting of cell-cycle proteins by miR-193a-3p is responsible for inhibition of proliferation and induction of apoptosis in cancer cells.

We also engineered lung cancer A549 cells and colon cancer LoVo cells to ectopically express cDNAs encoding cyclin D1, CDK2, and CDK6 (Figure S5G), which we found to represent targets of miR-214-5p (Table S1). In contrast to parental cells, where miR-214-5p blocked cell expansion, cancer cells expressing miRNA-resistant cell-cycle targets were refractory to the anti-proliferative effect of miR-214-5p (Figure 5E).

Given the observed strong effects of cell-cycle-targeting miRNAs on tumor cells, we examined the response of ten human non-transformed cell lines to the four miRNAs. We found that these miRNAs had a much milder effect on growth of non-transformed cells and that they did not trigger apoptosis (Figures 5F and 5G).

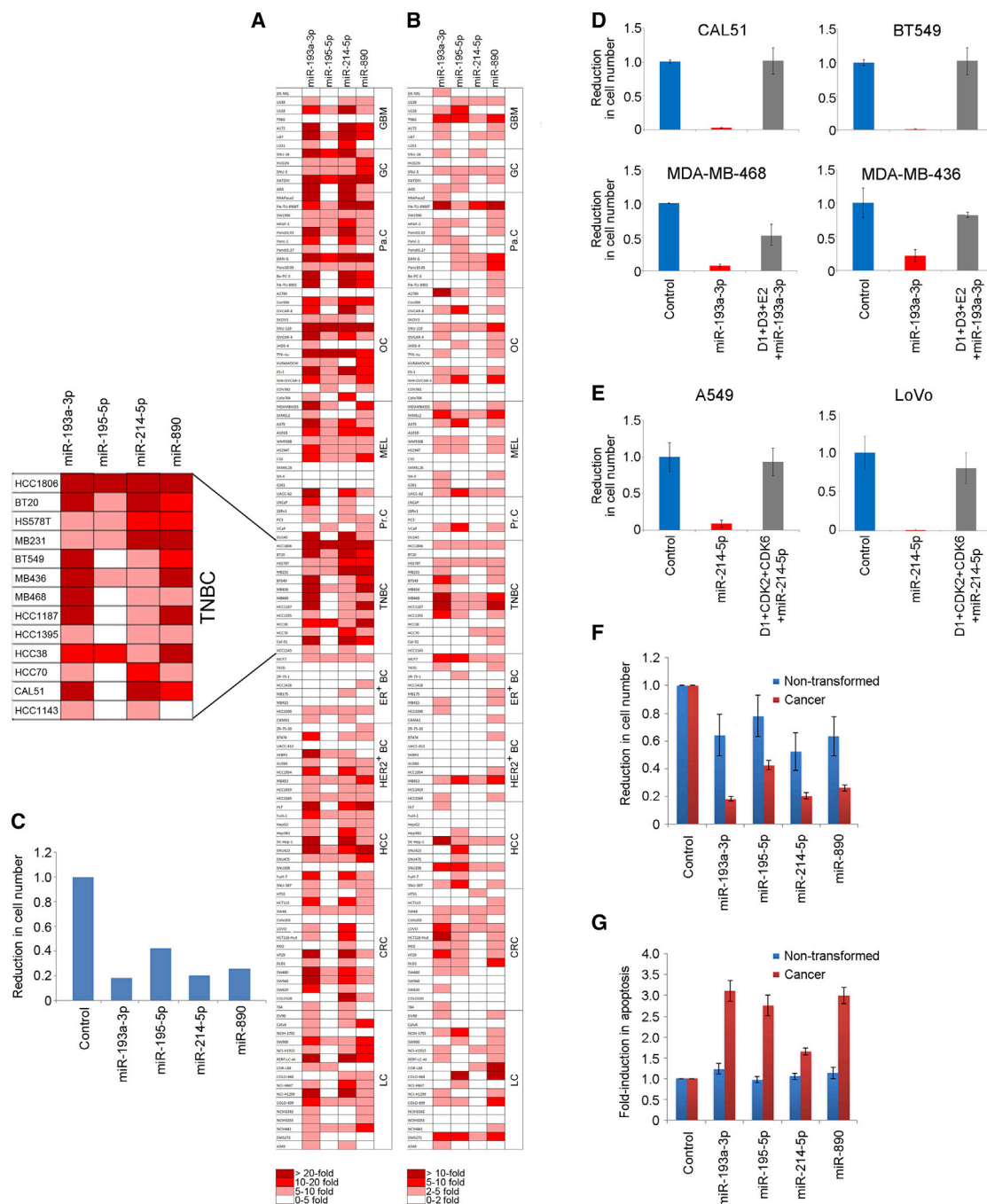


Figure 5. A Human Cancer Cell Line Encyclopedia Screen of Selected Cell-Cycle-Targeting miRNAs

(A) Four cell-cycle-targeting miRNAs (miR-193a-3p, miR-195-5p, miR-214-5p, miR-890), or a control miRNA were ectopically expressed in CCLE cell lines, and cell numbers were assessed after 6 days. The fold reduction of cell number expansion (compared with cells transfected with control miRNA) is indicated by the red color (see Table S4 for the mean raw values). The enlarged panel (left) shows the results from 13 triple-negative breast cancer lines. Each horizontal row corresponds to the indicated cell line, each vertical column to the indicated cell-cycle-targeting miRNA. For each miRNA and cell line, $n = 3$. Cell lines corresponding to the following cancer types were used: GBM, glioblastoma multiforme; GC, gastric cancer; Pa.C, pancreatic cancer; OC, ovarian cancer; MEL, melanoma; Pr.C, prostate cancer; TNBC, triple-negative breast cancer; ER⁺ BC, estrogen receptor-positive breast cancer; HER2⁺ BC, HER2-positive breast cancer; HCC, hepatocellular carcinoma; CRC, colorectal carcinoma; LC, lung carcinoma.

(B) Cells were monitored for apoptosis by a cleaved caspase-3 assay 24 hr post miRNA transfection. The fold-increase of cleaved caspase-3 (compared with cells transfected with control miRNA) is indicated by the red color. Data are presented as in (A). For each miRNA and cell line, $n = 3$.

(C) Average reduction in cell number expansion across all 122 CCLE cell lines for each of the four miRNAs tested. The reduction in cell number was defined as the ratio of cell numbers observed in cells transfected with this miRNA to cell numbers observed in cells transfected with control miRNA.

(legend continued on next page)

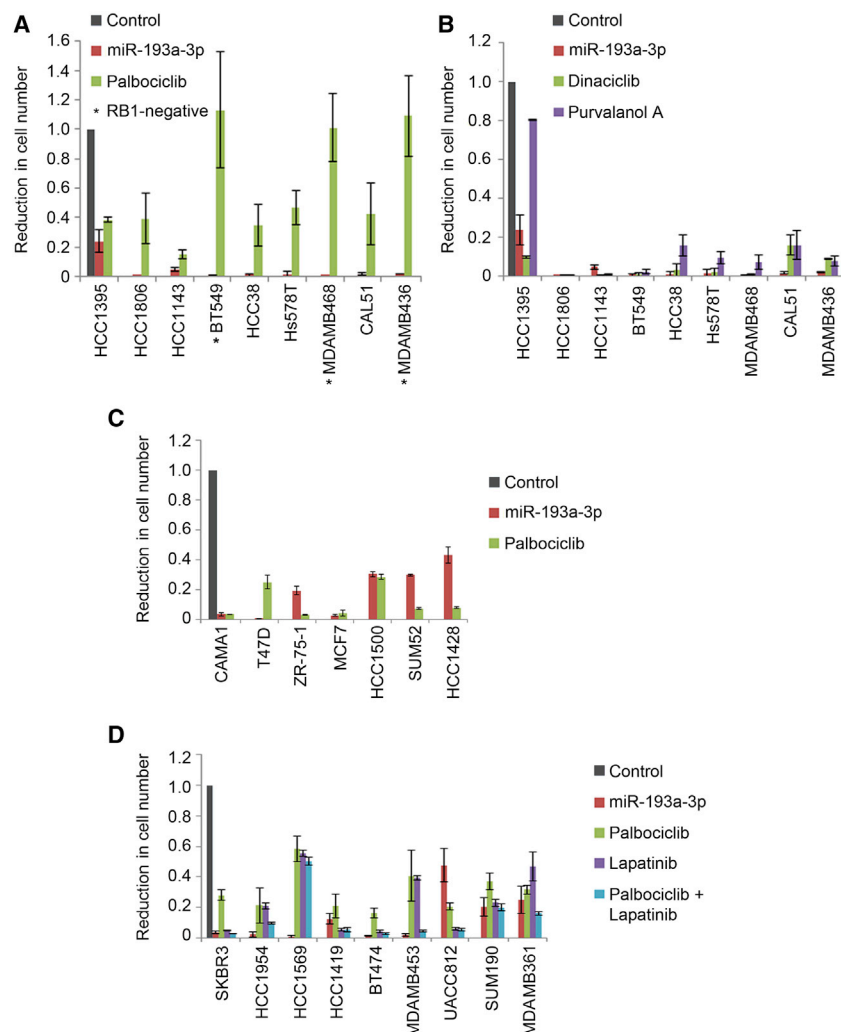


Figure 6. Comparison of the Effects of miR-193a-3p Versus Treatment of Breast Cancer Cells with CDK Inhibitors

(A) The indicated TNBC cell lines were transfected with miR-193a-3p or control miRNA, or treated with 1 μ M palbociclib or with DMSO (control). Cell numbers were determined after 6 days. *RB1-deleted cell lines.

(B) TNBC lines were transfected with miR-193a-3p or control miRNA, or treated with 10 nM dinaciclib or 10 μ M purvalanol A or DMSO (control). Cell numbers were determined after 6 days, as above.

(C) ER⁺ breast cancer cell lines were transfected or treated and monitored as in (A).

(D) HER2⁺ breast cancer cell lines were transfected with miR-193a-3p or control miRNA, or treated with 1 μ M palbociclib or 1 μ M lapatinib alone or in combination. Cell numbers were determined after 6 days.

In (A–D), for cells transfected with miR-193a-3p, the reduction in cell number expansion was defined as the ratio of cell numbers observed in cells transfected with this miRNA to cell numbers observed in cells transfected with control miRNA. For cells treated with chemical compounds, the reduction in cell number expansion was defined as the ratio of cell numbers observed in cells treated with this compound to cell numbers observed in cells transfected with DMSO. Black bar indicates that this ratio was 1.0 in control cells (transfected with control miRNA or DMSO-treated) and is shown only for one cell line. Data are mean values $\pm$ SD. For each bar, $n = 3$.

See also Figure S5.

Response of Human Breast Cancer Cells to miR-193a-3p Compared with Therapeutic Compounds

Our CCLE screen demonstrated that TNBC lines are highly sensitive to cell-cycle-targeting miRNAs and that miR-193a-3p had overall the strongest effect across cell lines representing all tumor types. In order to further evaluate the therapeutic potential of this miRNA, we compared the response of breast cancer cells to miR-193a-3p expression versus treatment with an inhibitor of

functional retinoblastoma protein, RB1 (Figure 6A). In contrast, ectopic expression of miR-193a-3p strongly decreased cell numbers in eight out of nine TNBC cell lines. Of particular interest, miR-193a-3p was potent against cell lines harboring deletion of the RB1 gene, pointing to the therapeutic potential of this miRNA against RB1-negative tumors (Figure 6A).

Chemical inhibition of CDKs was shown to have a profound effect on TNBC, as this tumor type may be particularly

(D) Reduction in cell number expansion in the indicated breast cancer cell lines transfected with control miRNA (blue bars) or miR-193a-3p (red bars). Gray bars, cells were engineered to express miRNA-resistant cyclins D1, D3, and E2 (D1 + D3 + E2), and transfected with control miRNA or with miR-193a-3p. Bars show the ratio of cell numbers observed in cells transfected with miR-193a-3p to cell numbers observed in cells transfected with control miRNA. Data are mean values $\pm$ SD. For each miRNA and cell line, $n = 3$.

(E) Similar analysis as in (D) for A549 and LoVo cells transfected with miR-214-5p. Gray bars show the reduction in cell number expansion upon miR-214-5p expression in cells engineered to express miRNA-resistant cyclin D1, CDK2, and CDK6 (D1 + CDK2 + CDK6), relative to cells transfected with control miRNA. Data are mean values $\pm$ SD. For each miRNA and cell line, $n = 3$.

(F) Reduction in cell number expansion in human non-transformed cell lines ($n = 10$, blue bars) and in cancer cells ($n = 122$, red bars) upon ectopic expression of the indicated cell-cycle-targeting miRNAs. Control, cells transfected with control miRNA. The reduction in cell number was defined as the ratio of cell numbers observed in cells transfected with this miRNA to cell numbers observed in cells transfected with control miRNA. Data are mean values $\pm$ SD. For each miRNA and cell line, $n = 3$.

(G) Fold induction of apoptosis in human non-transformed cell lines ($n = 10$, blue bars) and in cancer cells ($n = 122$, red bars) upon ectopic expression of the indicated cell-cycle-targeting miRNAs. Data are mean values $\pm$ SD. For each miRNA and cell line, $n = 3$.

See also Figure S5 and Table S4.

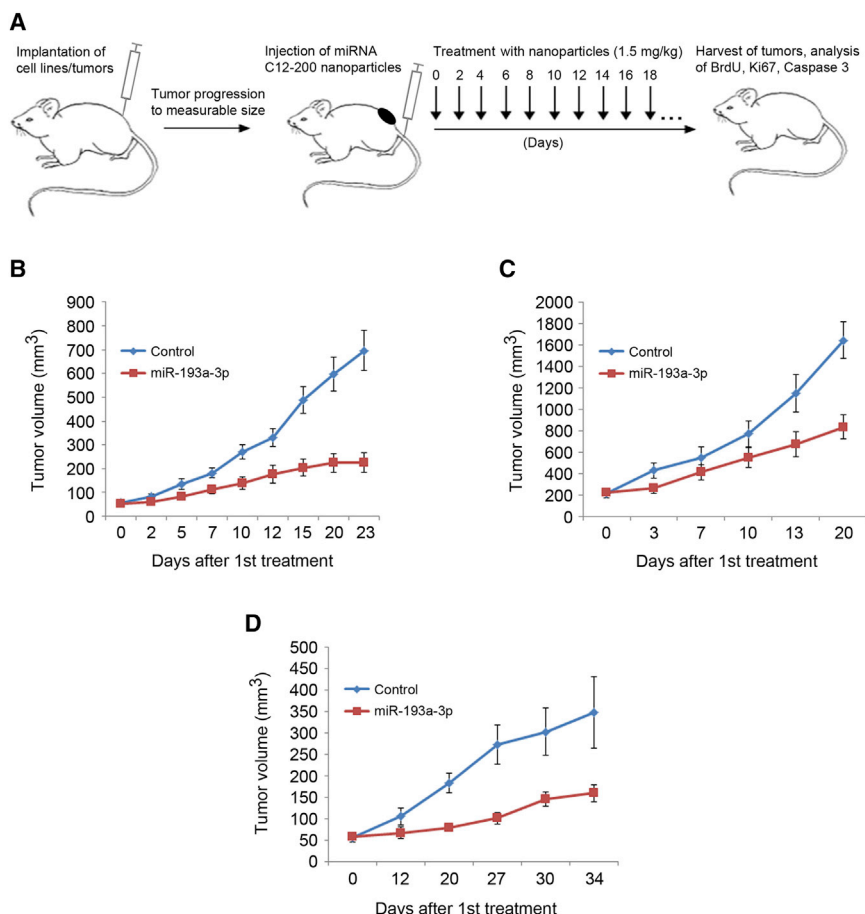


Figure 7. In Vivo Delivery of miR-193a-3p Inhibits Growth of Triple-Negative Breast Cancer Xenografts

(A) Outline of the experimental strategy. (B–D) Mice bearing xenografts derived from TNBC cell lines CAL51 (B) or HCC1806 (C) or bearing xenografts of a patient-derived TNBC (D) were systemically treated with nanoparticles containing miR-193a-3p or control miRNA. Shown are mean tumor volumes $\pm$ SEM; $n = 10$ per group. See also Figures S6 and S7.

Nanoparticle-Mediated Delivery of miR-193a-3p to Xenografts of Triple-Negative Breast Cancers

To extend these observations to an in vivo setting, we asked whether a systemic administration of miR-193a-3p to mice bearing xenografts of human TNBC would halt tumor progression. To deliver miR-193a-3p to tumor-bearing mice, we utilized nanoparticles based on epoxide-derived lipidoid C12-200. This type of nanoparticles was shown to avoid immune responses and to improve delivery of large bulky double-stranded nucleic acids to animals in vivo without causing toxicity (Love et al., 2010).

We first subcutaneously injected immunocompromised mice with TNBC CAL51 or HCC1806 cells leading to formation of tumors (see Figure S6A for the

dependent on CDK kinases for cancer cell survival (Horiuchi et al., 2012). We therefore compared the effects of miR-193a-3p expression in TNBC cells versus treatment with CDK inhibitors dinaciclib and purvalanol A, which inhibit CDK1, CDK2, and other cell-cycle kinases. We found that miR-193a-3p was generally equally or more potent than each of the two inhibitors (Figure 6B).

In order to explore the effect of miR-193a-3p on other types of breast cancers, we compared the response of human estrogen-receptor-positive (ER⁺) mammary carcinoma cell lines to miR-193a-3p versus palbociclib, which is currently being used in clinical trials for patients bearing ER⁺ tumors (Asghar et al., 2015; Finn et al., 2015). We found that miR-193a-3p was quite effective against ER⁺ breast cancer cell lines, although generally to a lesser extent than palbociclib (Figure 6C).

Lastly, we gauged the impact of miR-193a-3p expression on HER2⁺ breast cancer cell lines and compared it with the effect of lapatinib. Lapatinib, a dual HER2/EGFR tyrosine kinase inhibitor, is used for treatment of HER2⁺ breast cancers. We found that miR-193a-3p was equivalent to or superior to lapatinib alone in most cases and often had a comparable impact on cell number expansion with the combined effect of lapatinib plus palbociclib (Figure 6D). Together, these results suggest a therapeutic potential of miR-193a-3p for breast cancers and in particular for TNBCs.

endogenous levels of miR-193a-3p in these tumors). Once the animals developed palpable tumors, every 2–3 days we systematically administered nanoparticles containing miR-193a-3p or containing miRNA mimic negative control via tail vein injection (Figure 7A). We found that administration of nanoparticles with miR-193a-3p significantly inhibited tumor growth in vivo (Figures 7B and 7C). Analyses of tumors for 5-bromo-2-deoxyuridine (BrdU) uptake, Ki67 staining, and cleaved caspase-3 revealed significant reduction of proliferation and induction of apoptosis in miR-193a-3p-treated mice (Figures S6B–S6G).

To ascertain that the observed in vivo effects were mediated through the ability of miR-193a-3p to directly target cyclins/CDKs, we generated xenografts using CAL51 cells engineered to express miRNA-resistant cyclins D1, D3, and E2. Strikingly, administration of miR-193a-3p to tumor-bearing animals had essentially no inhibitory effect on tumor growth (Figure S6H). In contrast to xenografts derived from parental CAL51 cells, tumor cells expressing miRNA-resistant cyclins did not show reduced BrdU or Ki67 staining and were protected against miR-193a-3p-induced apoptosis (Figures S6I–S6K).

We further confirmed the anti-tumor effect in vivo using mice bearing orthotopic xenografts of a patient-derived TNBC. This tumor was refractory to conventional therapy and had progressed after treatment with doxorubicin and paclitaxel. Administration of miR-193a-3p to mice bearing this aggressive

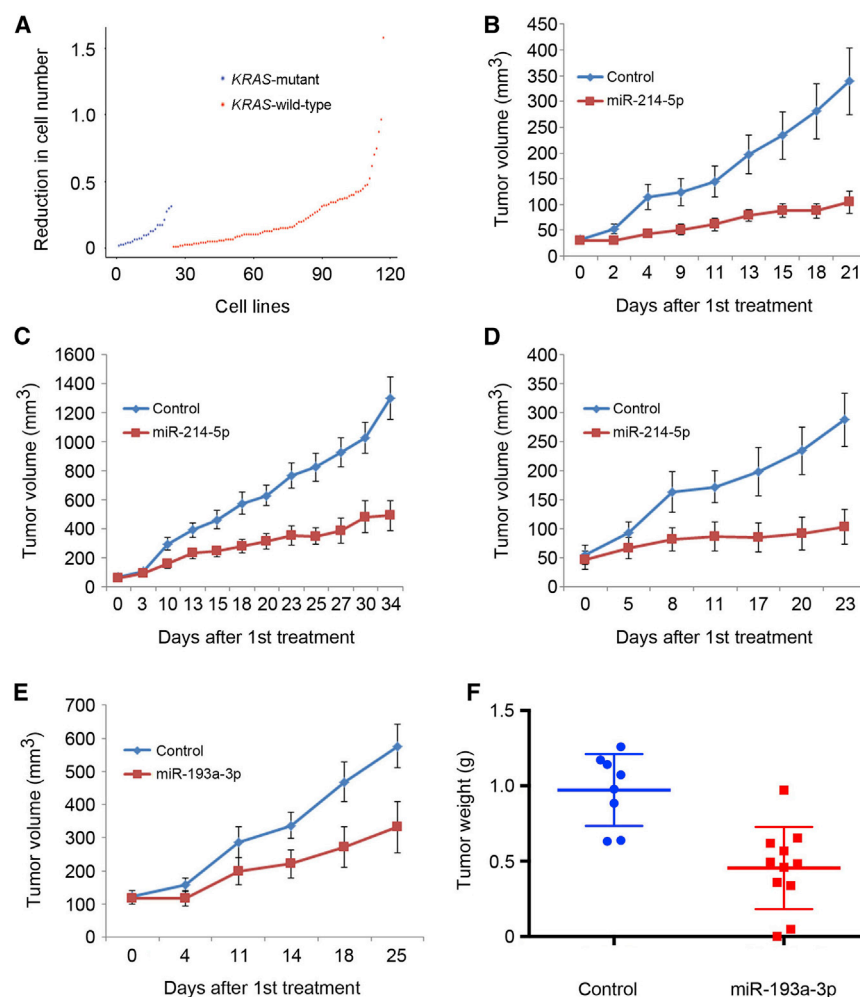


Figure 8. In Vivo Delivery of miR-214-5p or miR-193a-3p Inhibits Growth of Lung Cancer, Colon Cancer, and Sarcoma Xenografts

(A) A plot illustrating reduction in cell number expansion (vertical axis) upon expression of miR-214-5p in human cancer cell lines from CCLE carrying a *KRAS*-mutated allele (blue dots) or wild-type *KRAS* (red dots); each dot corresponds to a different cell line. The reduction in cell number expansion was defined as the ratio of cell numbers observed in cells transfected with this miRNA to cell numbers observed in cells transfected with control miRNA. Cell lines were arranged according to the decreasing inhibitory effect on cell number expansion. Statistics were performed using the Wilcoxon test.

(B–D) Tumor volumes (mean values $\pm$ SEM) in mice bearing xenografts derived from a *KRAS*-mutated lung adenocarcinoma cell line A549 (B), a *KRAS*-mutated colorectal cancer cell line LoVo (C), or a patient-derived metastatic *KRAS*-mutated colon adenocarcinoma (D), following systemic administration of nanoparticles containing miR-214-5p or containing control miRNA. In (B and C), $n = 10$ mice per group. In (D), $n = 8$ (control) and $n = 12$ (miR-214-5p).

(E) Tumor volumes (mean values $\pm$ SEM) in mice bearing xenografts of patient-derived dermatofibrosarcoma protuberans cells, following systemic administration of miR-193a-3p; $n = 8$ mice (control) and $n = 11$ (miR-193a-3p).

(F) Tumor weight measurements at day 25, from the experiment shown in (E). Each dot corresponds to a different tumor. Thick horizontal lines depict mean values; thin lines, SD.

See also Figures S7 and S8; Table S5.

tumor strongly inhibited tumor expansion (Figures 7D and S6L–S6N).

Analyses of nanoparticle-treated mice revealed that both normal organs as well as tumors displayed uptake of the exogenous miR-193a-3p (Figure S6O). Given this broad uptake, we tested the impact of long-term administration of miR-193a-3p on normal tissues. We found that continuous systemic administration of nanoparticles containing miR-193a-3p to animals had no detectable effect on proliferation rate in the internal organs (Figures S7A and S7B). Also, no increased apoptosis was observed in the internal organs of treated mice (Figure S7C). This is in contrast to tumor cells in nanoparticle-treated mice, where miR-193a-3p reduced proliferation and triggered apoptosis (Figures S6B–S6G and S6L–S6N).

We also analyzed the histopathological appearance of internal organs in mice treated long-term with miR-193a-3p-containing nanoparticles and found no apparent abnormalities (Figure S7D and data not shown).

Lastly, we determined that long-term administration of nanoparticles containing miR-193a-3p had no obvious effect on biochemical parameters in the peripheral blood of recipient mice (Figure S7E), no obvious effect on liver function (Figure S7F),

and no detectable effect on the function of the hematopoietic system (Figures S7G and S7H).

Collectively, these analyses indicate that nanoparticle-based delivery of miR-193a-3p may represent an efficient strategy against refractory TNBCs in vivo, by selectively targeting cancer cells while sparing normal, non-transformed tissues.

Nanoparticle-Mediated Delivery of miR-214-5p to *KRAS*-Mutant Cancers

Analyses of the response of 122 human cancer cell lines to the four cell-cycle-targeting miRNAs allowed us to correlate the observed response with genetic lesions carried by each of these cell lines, which have been annotated in CCLE. These analyses revealed that miR-214-5p is more potent against cell lines harboring mutations in *KRAS* than those with wild-type *KRAS* ($p = 0.02$ by Wilcoxon test) (Figure 8A).

We validated this finding in vivo using nanoparticle-mediated delivery of miR-214-5p to mice bearing xenografts of human *KRAS*-mutant lung cancer A549 cells or colon cancer LoVo cells (please see Figure S8A for the levels of miR-214-5p in these tumors). We found that a systemic administration of miR-214-5p inhibited growth of *KRAS*-mutant tumors in vivo (Figures 8B, 8C, and S8B–S8E).

We extended these observations using mice bearing xenografts of patient-derived metastatic *KRAS*-mutant colon adenocarcinoma. The primary tumor was refractory to treatments with FOLFOX (folinic acid + 5-fluorouracil + oxaliplatin) and FOLFIRI (folinic acid + 5-fluorouracil + irinotecan) + bevacizumab. Administration of miR-214-5p to mice bearing this aggressive cancer strongly inhibited tumor growth (Figures 8D, S8F, and S8G).

As was the case for miR-193a-3p, repeated administration of miR-214-5p to mice had no detectable effect on cell proliferation in the internal organs (Figures S7A and S7B). The recipient mice displayed unperturbed histopathological appearance of the internal organs (Figure S7D), and normal blood chemistry, liver function and hematological parameters (Figures S7E–S7H).

Collectively, these observations suggest that nanoparticle-based delivery of miR-214-5p might represent an effective therapeutic strategy against *KRAS*-mutant tumors, including aggressive, treatment-refractory cases.

To exclude a possibility that our miRNA anti-tumor effects might have been caused by a non-specific activation of the immune system, we monitored liver expression of a panel of cytokines, 2 days post injection of miR-214-5p. We detected no major changes in the cytokine levels, essentially ruling out a non-specific immune response as the underlying mechanism (Figure S8H).

Expression-Based Algorithm to Predict the Response of Tumors to Cell-Cycle-Targeting miRNAs

By correlating gene expression profiles of each of the 122 human cancer cell lines available from CCLE with the response to the four cell-cycle-targeting miRNAs, we developed a high-dimensional regression method based on elastic net (Zou and Hastie, 2005), which predicts the response of cancer cells to each of the miRNAs (Table S5). In order to test whether this algorithm is applicable to patient-derived tumors, we analyzed gene expression of an aggressive dermatofibrosarcoma protuberans that was refractory to conventional therapy and progressed despite treatment with imatinib (Gleevec) and PI3K inhibitor pictilisib (GDC-0941). By querying gene expression data with our algorithm, we predicted that this tumor would be responsive to miR-193a-3p (Table S5). To validate this prediction in vivo, we implanted patient-derived tumor cells into immunocompromised mice and systematically delivered nanoparticles containing miR-193a-3p. Indeed, administration of miR-193a-3p significantly inhibited tumor growth in vivo (Figures 8E, 8F, and S8I–S8K). These observations suggest that our expression-based algorithm allows one to predict the response of primary tumors to individual miRNAs.

DISCUSSION

A role for miRNAs in the pathogenesis of cancer was suggested shortly after their discovery in humans 16 years ago (Lagos-Quintana et al., 2001). The study of Lu et al. (2005) revealed that miRNAs are generally downregulated in human cancers, suggesting that many miRNAs might play tumor-suppressive roles. Indeed, reduced expression of the let-7 family of miRNAs was associated with clinical progression of lung cancers, while the loss of miR-15/16 was linked to progression in chronic lymphocytic leukemia (Calin et al., 2005; Takamizawa et al., 2004). However, very few individual miRNAs have been validated to play tumor-suppressive roles using genetic systems. This might

seem surprising in the light of numerous reports linking miRNAs to cancer but could be explained by the highly redundant function of these small regulatory RNAs in physiological settings (Bartel, 2009).

Several groups proposed to use miRNA overexpression systems as an anti-cancer therapeutic strategy, and this approach led to promising outcomes in vivo (Kota et al., 2009; Liu et al., 2011). In particular, a synthetic miR-34a mimic is currently in clinical trials (NCT01829971) for treatment of patients with primary liver cancer or with liver metastases. Interestingly, miRNAs used in these in vivo systems have been reported to regulate, among other targets, also cyclins and/or CDKs, highlighting a potential link between successful miRNA mimic therapy and repression of the cell-cycle machinery.

To explore this possible therapeutic link, we performed nine genome-wide screens to enumerate miRNAs that directly regulate the cell-cycle machinery. We uncovered a class of miRNAs in the human genome, which target all, or nearly all cyclins and CDKs. We showed that these cell-cycle-targeting miRNAs effectively inhibit proliferation of a wide array of cancer cell types. We demonstrated that these miRNAs can be systematically administered to animals in the form of nanoparticles where they effectively inhibit proliferation of tumor cells while sparing normal, non-transformed tissues. We extended our studies to patient-derived xenografts of tumors that were refractory to standard treatments and demonstrated the efficacy of using cell-cycle-targeting miRNAs to target these incurable tumors. Hence, with improved delivery methods to tumor tissues, these miRNAs might offer an effective anti-cancer strategy against deadly, refractory cancer types.

In addition to inducing proliferative arrest, cell-cycle-targeting miRNAs also triggered cell death of several cancer cell lines. This observation is in agreement with published reports that inhibition of cell-cycle kinases or downregulation of cyclins can trigger tumor cell apoptosis, in addition to cell-cycle arrest (Choi et al., 2012; Goga et al., 2007; Horiuchi et al., 2012; Molenaar et al., 2009; Sawai et al., 2012).

It is not clear why ectopic expression of the four cell-cycle-targeting miRNAs had the strongest effect on TNBCs and gastric cancers. One possible explanation is that TNBCs were shown to be particularly dependent on CDKs for cancer cell survival (Horiuchi et al., 2012). Consistent with this hypothesis, expression of cell-cycle-targeting miRNAs triggered apoptosis of TNBC cells in vitro and in vivo, and this response was prevented by ectopic expression of miRNA-resistant cell-cycle targets in cancer cells.

Our analyses revealed that a higher proportion of *KRAS*-mutant cancer cell lines was potently inhibited by miR-214-5p, compared with cancer cells expressing wild-type *KRAS*. miR-214-5p is not predicted to target *KRAS* itself, and expression of miRNA-resistant cell-cycle targets (cyclin D1, CDK2, and CDK6) prevented the anti-proliferative effect of this miRNA. Of note, high-throughput screens have previously uncovered synthetic lethal interactions between oncogenic *KRAS* and inhibition of several CDKs (Barbie et al., 2009). Moreover, *KRAS*-mutant tumors were shown to be more sensitive to CDK inhibition than *KRAS* wild-type tumors (Macias et al., 2007; Puyol et al., 2010), which might explain their increased sensitivity to miR-214-5p.

Our observation that cell-cycle-targeting miRNAs affected cancer cells while sparing non-transformed tissues was not

entirely unexpected. Several studies demonstrated that cancer cells are particularly dependent on cyclins and CDKs for their proliferation and survival, and hence inhibition of these proteins is predicted to preferentially affect cancer cells (Campaner et al., 2010; Choi et al., 2012; Puyol et al., 2010). Moreover, expression of miRNAs decreases, but does not fully extinguish, expression of their targets, and such a global dampening of cell-cycle machinery is likely to especially affect transformed cells.

While our studies mostly explored the effects of overexpression of cell-cycle-targeting miRNAs, it might be expected that at least some of them normally play growth-suppressive roles when expressed at physiological levels. Consistent with this notion, our analyses revealed that the gene encoding miR-193a-3p is frequently deleted in human cancers. miR-193a-3p was previously proposed to play tumor-suppressive functions by repressing EGFR signaling (Uhlmann et al., 2012), by promoting apoptosis via repression of MCL1 (Kwon et al., 2013; Williams et al., 2015), and by suppressing tumor cell migration and invasion through small GTPase Rab27B (Pu et al., 2016) or ERBB4 and S6K2 (Liang et al., 2015; Yu et al., 2015). The loss of these anti-tumor functions of miR-193a-3p may contribute to the neoplastic transformation in cancers harboring deletion of the miR-193a locus.

Another cell-cycle-targeting miRNA highlighted by our study is miR-195-5p. We found that this miRNA might display growth-suppressive properties *in vivo*, as demonstrated by a significant downregulation of miR-195-5p in 11 types of human tumors (Figure 3B). Consistent with our findings, downregulation of miR-195-5p was previously reported in human breast and colorectal cancers (Chen et al., 2015; Luo et al., 2014, 2016).

By correlating gene expression profiles of human cancer cell lines versus the responses to individual miRNA, we generated an expression-based algorithm that can predict the response of a particular tumor to a given miRNA. In the future, this approach can be extended to examine the effects of all miRNAs. By applying similar algorithms to the one described in our study, one will be able to identify a miRNA, or a combination thereof, that would be particularly potent against a given individual human tumor. This approach might lead to highly individualized miRNA-based anti-cancer therapies.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

- KEY RESOURCES TABLE
- CONTACT FOR REAGENT AND RESOURCE SHARING
- EXPERIMENTAL MODEL AND SUBJECT DETAILS
 - Animals
 - Cell Lines
 - Cell Line Authentication
- METHOD DETAILS
 - 3'UTR Cloning Strategy and miR-193a PCR
 - Screening of miRNA Library
 - Mutant miRNA Mimics, Mutant Cyclin/CDK 3'UTRs
 - Cells Expressing miRNA-Resistant Cyclin/CDKs
 - Cell Cycle Synchronization

- Apoptosis and Senescence Assays
- Analyses of Cultured Cancer Cell Lines
- CCLE Cell Line Screen
- Western Blotting
- miRNA Transfection and miRNA qPCR
- RT-qPCR Analysis
- Flow Cytometry Analysis
- Formulation of miRNAs into Nanoparticles
- Nanoparticle-Mediated Delivery of miRNAs
- Analyses of miRNA Uptake *In Vivo*
- Analyses of Liver Cytokine mRNA Levels
- Embedding, Sectioning and Staining
- Hematological Analyses
- Computational Analyses

● QUANTIFICATION AND STATISTICAL ANALYSIS

- Quantification of IHC Staining
- Statistical Analysis

● DATA AND SOFTWARE AVAILABILITY

- Deposited Data

SUPPLEMENTAL INFORMATION

Supplemental Information includes eight figures and seven tables and can be found with this article online at <http://dx.doi.org/10.1016/j.ccell.2017.03.004>.

AUTHOR CONTRIBUTIONS

P.H. and P.S. designed the study and wrote the paper. P.H. performed the experiments. Y.W., X.L., C.J., and Y.L. contributed computational analyses, which were directed and supervised by C.L. A.F., V.M., T.O., Y.J.C., K.E.S., J.M.S., and D.F. helped with experiments. L.A. helped with experiments and with screen design. H. Yin, R.L.B., K.J.K., J.Y., and D.G.A. provided nanoparticle-formulated-miRNAs. S.G., H. Yuzugullu, and J.J.Z. helped with experiments and contributed xenografts of TNBC. R.S. provided colon cancer specimens, K.N. xenografts of colon cancer, E.S. xenografts of colon cancer and dermatofibrosarcoma. K.P. helped with designing breast cancer studies. P.S. directed the study.

ACKNOWLEDGMENTS

This was supported by NIH grants 5 R01 CA083688, 5 R01 CA132740, and 5 P01 CA080111 (to P.S.). P.H. was partly supported by a post-doctoral fellowship from the Wennergren Foundations (Stockholm, Sweden), Y.W. and C.L. by grants from Peking-Tsinghua Center for Life Sciences and National Natural Science Foundation of China Key Research Grant 71532001, J.M.S. by an MNISW Fellowship Mobilność Plus. P.S. and K.P. are consultants and recipients of research grants from Novartis. We thank Dr. Caroline Shamu from ICCB-Longwood Screening Facility for the miRNA mimic library and help with designing the screen.

Received: January 22, 2016

Revised: December 6, 2016

Accepted: March 9, 2017

Published: April 10, 2017

REFERENCES

- Adams, B.D., Kasinski, A.L., and Slack, F.J. (2014). Aberrant regulation and function of microRNAs in cancer. *Curr. Biol.* 24, R762–R776.
- Asghar, U., Witkiewicz, A.K., Turner, N.C., and Knudsen, E.S. (2015). The history and future of targeting cyclin-dependent kinases in cancer therapy. *Nat. Rev. Drug Discov.* 14, 130–146.
- Barbie, D.A., Tamayo, P., Boehm, J.S., Kim, S.Y., Moody, S.E., Dunn, I.F., Schinzel, A.C., Sandy, P., Meylan, E., Scholl, C., et al. (2009). Systematic

- RNA interference reveals that oncogenic KRAS-driven cancers require TBK1. *Nature* 462, 108–112.
- Bartel, D.P. (2009). MicroRNAs: target recognition and regulatory functions. *Cell* 136, 215–233.
- Beroukhi, R., Mermel, C.H., Porter, D., Wei, G., Raychaudhuri, S., Donovan, J., Barretina, J., Boehm, J.S., Dobson, J., Urashima, M., et al. (2010). The landscape of somatic copy-number alteration across human cancers. *Nature* 463, 899–905.
- Bonci, D., Coppola, V., Musumeci, M., Addario, A., Giuffrida, R., Memeo, L., D'Urso, L., Pagliuca, A., Biffoni, M., Labbaye, C., et al. (2008). The miR-15a-miR-16-1 cluster controls prostate cancer by targeting multiple oncogenic activities. *Nat. Med.* 14, 1271–1277.
- Calin, G.A., Ferracin, M., Cimmino, A., Di Leva, G., Shimizu, M., Wojcik, S.E., Iorio, M.V., Visone, R., Sever, N.I., Fabbri, M., et al. (2005). A MicroRNA signature associated with prognosis and progression in chronic lymphocytic leukemia. *N. Engl. J. Med.* 353, 1793–1801.
- Campaner, S., Doni, M., Hydring, P., Verrecchia, A., Bianchi, L., Sardella, D., Schleker, T., Perna, D., Tronnersjö, S., Murga, M., et al. (2010). Cdk2 suppresses cellular senescence induced by the c-myc oncogene. *Nat. Cell Biol.* 12, 54–59, sup pp 1–14.
- Chen, X., Shi, K., Wang, Y., Song, M., Zhou, W., Tu, H., and Lin, Z. (2015). Clinical value of integrated-signature miRNAs in colorectal cancer: miRNA expression profiling analysis and experimental validation. *Oncotarget* 6, 37544–37556.
- Choi, Y.J., Li, X., Hydring, P., Sanda, T., Stefano, J., Christie, A.L., Signoretti, S., Look, A.T., Kung, A.L., von Boehmer, H., and Sicinski, P. (2012). The requirement for cyclin D function in tumor maintenance. *Cancer Cell* 22, 438–451.
- Finn, R.S., Dering, J., Conklin, D., Kalous, O., Cohen, D.J., Desai, A.J., Ginther, C., Atefi, M., Chen, I., Fowst, C., et al. (2009). PD 0332991, a selective cyclin D kinase 4/6 inhibitor, preferentially inhibits proliferation of luminal estrogen receptor-positive human breast cancer cell lines in vitro. *Breast Cancer Res.* 11, R77.
- Finn, R.S., Crown, J.P., Lang, I., Boer, K., Bondarenko, I.M., Kulyk, S.O., Ettl, J., Patel, R., Pinter, T., Schmidt, M., et al. (2015). The cyclin-dependent kinase 4/6 inhibitor palbociclib in combination with letrozole versus letrozole alone as first-line treatment of oestrogen receptor-positive, HER2-negative, advanced breast cancer (PALOMA-1/TRIO-18): a randomised phase 2 study. *Lancet Oncol.* 16, 25–35.
- Goga, A., Yang, D., Tward, A.D., Morgan, D.O., and Bishop, J.M. (2007). Inhibition of CDK1 as a potential therapy for tumors over-expressing MYC. *Nat. Med.* 13, 820–827.
- Hayes, J., Peruzzi, P.P., and Lawler, S. (2014). MicroRNAs in cancer: biomarkers, functions and therapy. *Trends Mol. Med.* 20, 460–469.
- Horiuchi, D., Kusdra, L., Huskey, N.E., Chandriani, S., Lenburg, M.E., Gonzalez-Angulo, A.M., Creasman, K.J., Bazarov, A.V., Smyth, J.W., Davis, S.E., et al. (2012). MYC pathway activation in triple-negative breast cancer is synthetic lethal with CDK inhibition. *J. Exp. Med.* 209, 679–696.
- Johnson, C.D., Esquela-Kerscher, A., Stefani, G., Byrom, M., Kelnar, K., Ovcharenko, D., Wilson, M., Wang, X., Shelton, J., Shingara, J., et al. (2007). The let-7 microRNA represses cell proliferation pathways in human cells. *Cancer Res.* 67, 7713–7722.
- Kota, J., Chivukula, R.R., O'Donnell, K.A., Wentzel, E.A., Montgomery, C.L., Hwang, H.W., Chang, T.C., Vivekanandan, P., Torbenson, M., Clark, K.R., et al. (2009). Therapeutic microRNA delivery suppresses tumorigenesis in a murine liver cancer model. *Cell* 137, 1005–1017.
- Kwon, J.E., Kim, B.Y., Kwak, S.Y., Bae, I.H., and Han, Y.H. (2013). Ionizing radiation-inducible microRNA miR-193a-3p induces apoptosis by directly targeting Mcl-1. *Apoptosis* 18, 896–909.
- Lagos-Quintana, M., Rauhut, R., Lendeckel, W., and Tuschl, T. (2001). Identification of novel genes coding for small expressed RNAs. *Science* 294, 853–858.
- Liang, H., Liu, M., Yan, X., Zhou, Y., Wang, W., Wang, X., Fu, Z., Wang, N., Zhang, S., Wang, Y., et al. (2015). miR-193a-3p functions as a tumor suppressor in lung cancer by down-regulating ERBB4. *J. Biol. Chem.* 290, 926–940.
- Liu, C., Kelnar, K., Liu, B., Chen, X., Calhoun-Davis, T., Li, H., Patrawala, L., Yan, H., Jeter, C., Honorio, S., et al. (2011). The microRNA miR-34a inhibits prostate cancer stem cells and metastasis by directly repressing CD44. *Nat. Med.* 17, 211–215.
- Love, K.T., Mahon, K.P., Levins, C.G., Whitehead, K.A., Querbes, W., Dorkin, J.R., Qin, J., Cantley, W., Qin, L.L., Racie, T., et al. (2010). Lipid-like materials for low-dose, in vivo gene silencing. *Proc. Natl. Acad. Sci. USA* 107, 1864–1869.
- Lu, J., Getz, G., Miska, E.A., Alvarez-Saavedra, E., Lamb, J., Peck, D., Sweet-Cordero, A., Ebert, B.L., Mak, R.H., Ferrando, A.A., et al. (2005). MicroRNA expression profiles classify human cancers. *Nature* 435, 834–838.
- Luo, Q., Wei, C., Li, X., Li, J., Chen, L., Huang, Y., Song, H., Li, D., and Fang, L. (2014). MicroRNA-195-5p is a potential diagnostic and therapeutic target for breast cancer. *Oncol. Rep.* 31, 1096–1102.
- Luo, Q., Zhang, Z., Dai, Z., Basnet, S., Li, S., Xu, B., and Ge, H. (2016). Tumor-suppressive microRNA-195-5p regulates cell growth and inhibits cell cycle by targeting cyclin dependent kinase 8 in colon cancer. *Am. J. Transl. Res.* 8, 2088–2096.
- Macias, E., Kim, Y., Miliani de Marval, P.L., Klein-Szanto, A., and Rodriguez-Puebla, M.L. (2007). Cdk2 deficiency decreases ras/CDK4-dependent malignant progression, but not myc-induced tumorigenesis. *Cancer Res.* 67, 9713–9720.
- Malumbres, M., and Barbacid, M. (2009). Cell cycle, CDKs and cancer: a changing paradigm. *Nat. Rev. Cancer* 9, 153–166.
- Molenaar, J.J., Ebus, M.E., Geerts, D., Koster, J., Lamers, F., Valentijn, L.J., Westerhout, E.M., Versteeg, R., and Caron, H.N. (2009). Inactivation of CDK2 is synthetically lethal to MYCN over-expressing cancer cells. *Proc. Natl. Acad. Sci. USA* 106, 12968–12973.
- Musgrove, E.A., Caldon, C.E., Barraclough, J., Stone, A., and Sutherland, R.L. (2011). Cyclin D as a therapeutic target in cancer. *Nat. Rev. Cancer* 11, 558–572.
- Pu, Y., Zhao, F., Cai, W., Meng, X., Li, Y., and Cai, S. (2016). MiR-193a-3p and miR-193a-5p suppress the metastasis of human osteosarcoma cells by down-regulating Rab27B and SRR, respectively. *Clin. Exp. Metastasis* 33, 359–372.
- Puyol, M., Martin, A., Dubus, P., Mulero, F., Pizcueta, P., Khan, G., Guerra, C., Santamaria, D., and Barbacid, M. (2010). A synthetic lethal interaction between K-Ras oncogenes and Cdk4 unveils a therapeutic strategy for non-small cell lung carcinoma. *Cancer Cell* 18, 63–73.
- Sawai, C.M., Freund, J., Oh, P., Ndiaye-Lobry, D., Bretz, J.C., Strikoudis, A., Genesca, L., Trimarchi, T., Kelliher, M.A., Clark, M., et al. (2012). Therapeutic targeting of the cyclin D3:CDK4/6 complex in T cell leukemia. *Cancer Cell* 22, 452–465.
- Takamizawa, J., Konishi, H., Yanagisawa, K., Tomida, S., Osada, H., Endoh, H., Harano, T., Yatabe, Y., Nagino, M., Nimura, Y., et al. (2004). Reduced expression of the let-7 microRNAs in human lung cancers in association with shortened postoperative survival. *Cancer Res.* 64, 3753–3756.
- Uhlmann, S., Mannsperger, H., Zhang, J.D., Horvat, E.A., Schmidt, C., Kublbeck, M., Henjes, F., Ward, A., Tschulena, U., Zweig, K., et al. (2012). Global microRNA level regulation of EGFR-driven cell-cycle protein network in breast cancer. *Mol. Syst. Biol.* 8, 570.
- Williams, M., Kirschner, M.B., Cheng, Y.Y., Hanh, J., Weiss, J., Mugridge, N., Wright, C.M., Linton, A., Kao, S.C., Edelman, J.J., et al. (2015). miR-193a-3p is a potential tumor suppressor in malignant pleural mesothelioma. *Oncotarget* 6, 23480–23495.
- Yu, T., Li, J., Yan, M., Liu, L., Lin, H., Zhao, F., Sun, L., Zhang, Y., Cui, Y., Zhang, F., et al. (2015). MicroRNA-193a-3p and -5p suppress the metastasis of human non-small-cell lung cancer by downregulating the ERBB4/PIK3R3/mTOR/S6K2 signaling pathway. *Oncogene* 34, 413–423.
- Zou, H., and Hastie, T. (2005). Regularization and variable selection via the elastic net. *J. R. Stat. Soc. B* 67 (Part 2), 301–320.

STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Cyclin D1 (rabbit monoclonal)	Millipore	Cat # 04-1151; RRID: AB_10615820
Cyclin D3 (rabbit polyclonal)	Santa Cruz Biotechnology	Cat # sc-182; RRID: AB_2259653
Cyclin E1 (mouse monoclonal)	Santa Cruz Biotechnology	Cat # sc-247; RRID: AB_627357
Cyclin E2 (rabbit monoclonal)	Millipore	Cat # 04-223; RRID: AB_1586967
Cyclin A2 (mouse monoclonal)	Sigma-Aldrich	Cat # MABC205; RRID: AB_11204171
Cyclin B1 (rabbit polyclonal)	Abcam	Cat # ab7957; RRID: AB_2072263
CDK1 (rabbit monoclonal)	Abcam	Cat # ab133327; RRID: AB_11155333
CDK2 (mouse monoclonal)	Santa Cruz Biotechnology	Cat # sc-6248; RRID: AB_627238
CDK4 (rabbit polyclonal)	Santa Cruz Biotechnology	Cat # sc-260; RRID: AB_631219
CDK6 (mouse monoclonal)	Abcam	Cat # ab54576; RRID: AB_940952
GAPDH (rabbit polyclonal)	Abcam	Cat # ab9485; RRID: AB_307275
Beta-Actin (mouse monoclonal)	Sigma-Aldrich	Cat # A5441; RRID: AB_476744
BrdU-FITC	BD Biosciences	Cat # 557891
Chemicals, Peptides, DNA and Recombinant Proteins		
DMSO	Sigma-Aldrich	Cat # D2650
Palbociclib	Medchem Express, NJ	Cat # HY-50767A
Dinaciclib	Chemietek	Cat # CT-DINA
BrdU	Sigma-Aldrich	Cat # B5002-1G
G418	Sigma-Aldrich	Cat # A1720
Thymidine	Sigma-Aldrich	Cat # T9250
Purvalanol A	Sigma-Aldrich	Cat # P4484
Lapatinib	Sigma-Aldrich	Cat # CDS022971
Propidium iodide	Sigma-Aldrich	Cat # P4170
Epidermal Growth Factor (EGF)	Peptotech	Cat # AF-100-15
Sodium Pyruvate (NaP)	Sigma-Aldrich	Cat # P5280
Non-Essential Amino Acids (NEAA)	Sigma-Aldrich	Cat # M7145
Hydrocortisone	Sigma-Aldrich	Cat # H0888
Cholera toxin	Sigma-Aldrich	Cat # C8052
Insulin	Sigma-Aldrich	Cat # I1882
Glutathione	Sigma-Aldrich	Cat # G6013
Human genomic DNA	Promega	Cat # G3041
Critical Commercial Assays		
Lipofectamine RNAiMAX transfection reagent	Thermo Fisher Scientific	Cat # 13778150
Lipofectamine 2000 transfection reagent	Thermo Fisher Scientific	Cat # 11668019
QuikChange II XL Site-Directed Mutagenesis	Agilent Technologies	Cat # 200522
Dual-Glo luciferase assay system	Promega	Cat # E2940
Senescence detection kit	Biovision	Cat # K320-250
Caspase-Glo 3/7 Assay	Promega	Cat # G8091
100 bp DNA ladder	New England Biolabs	Cat # N3231S
Matrigel	Corning	Cat # 354248
mirVana miRNA isolation kit, with phenol	Thermo Fisher Scientific	Cat # AM1560
GeneChip Human Genome U133 Plus 2.0 Array	Affymetrix	Cat # 900466

(Continued on next page)

Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
High Capacity cDNA Reverse Transcription Kit	Applied Biosystems	Cat # 4374966
Power SYBR® Green PCR Master Mix	Applied Biosystems	Cat # 4367659
TaqMan® MicroRNA Assays for miR-193a-3p	Applied Biosystems	Cat # 4427975, ID # 002250
TaqMan® MicroRNA Assays for miR-214-5p	Applied Biosystems	Cat # 4427975, ID # 002293
TaqMan® MicroRNA Control Assays for U6 snRNA	Applied Biosystems	Cat # 4427975, ID # 001973
Cell Culture Medium		
DMEM	Thermo Fisher Scientific	Cat # 41965-062
McCoy's 5A	ATCC	Cat # 30-2007
EMEM	ATCC	Cat # 30-2003
RPMI	Thermo Fisher Scientific	Cat # 11875-119
F-12K	Thermo Fisher Scientific	Cat # 21127022
L15	ATCC	Cat # 30-2008
DMEM:F12	Thermo Fisher Scientific	Cat # 31331093
SUM	Polyak lab / DFCI	NA
Waymouth's mb752/1	Thermo Fisher Scientific	Cat # 11220035
MEM	Thermo Fisher Scientific	Cat # 11095080
Opti-MEM I	Thermo Fisher Scientific	Cat # 31985070
MEGM	Lonza	Cat # CC-3150
BEGM	Lonza	Cat # CC-3170
DCBM	ATCC	Cat # PCS-200-030
Deposited Data		
Gene expression array	This paper	GEO: GSE77228
RNA-seq	This paper	GEO: GSE77229
Merged file of GSE77228 and GSE77229	This paper	GEO: GSE77230
Experimental Models: Cell Lines: See Table S6		
Experimental Models: Organisms/Strains		
NCRNU females	Taconic	CrTac:NCR-Foxn1<nu>
NOG females	Taconic	NOD.Cg-Prkdc<scid>Il2rg/JicTac
U2OS-pmirGLO-Cyclin D1-3'UTR	This paper	NA
U2OS-pmirGLO-Cyclin D2-3'UTR	This paper	NA
U2OS-pmirGLO-Cyclin D3-3'UTR	This paper	NA
U2OS-pmirGLO-Cyclin E1-3'UTR	This paper	NA
U2OS-pmirGLO-Cyclin E2-3'UTR	This paper	NA
U2OS-pmirGLO-CDK1-3'UTR	This paper	NA
U2OS-pmirGLO-CDK2-3'UTR	This paper	NA
U2OS-pmirGLO-CDK4-3'UTR	This paper	NA
U2OS-pmirGLO-CDK6-3'UTR	This paper	NA
Recombinant DNA		
pmirGLO-Cyclin D1-3'UTR	This paper	NA
pmirGLO-Cyclin D2-3'UTR	This paper	NA
pmirGLO-Cyclin D3-3'UTR	This paper	NA
pmirGLO-Cyclin E1-3'UTR	This paper	NA
pmirGLO-Cyclin E2-3'UTR	This paper	NA
pmirGLO-Cyclin A2-3'UTR	This paper	NA
pmirGLO-Cyclin B1-3'UTR	This paper	NA

(Continued on next page)

Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
pmirGLO-Cyclin B2-3'UTR	This paper	NA
pmirGLO-CDK1-3'UTR	This paper	NA
pmirGLO-CDK2-3'UTR	This paper	NA
pmirGLO-CDK4-3'UTR	This paper	NA
pmirGLO-CDK6-3'UTR	This paper	NA
pCMV-Cyclin D1	This paper	NA
pCMV-Cyclin D3	This paper	NA
pCMV-Cyclin E2	This paper	NA
pCAG-CDK2-CDK6	This paper	NA

Sequence-Based Reagents: See [Table S7](#)

Software and Algorithms

TargetScan Human Release 6.2	http://www.targetscan.org/vert_61/	NA
miRWalk2.0	http://zmf.umm.uni-heidelberg.de/apps/zmf/mirwalk2/	NA
miRanda	http://www.miRNA.org/miRNA/getMirnaForm.do	NA
miRDB	http://mirdb.org/cgi-bin/search.cgi	NA
TargetSpy	http://webclu.bio.wzw.tum.de/targetspy/index.php	NA
Ensembl release 87	http://www.ensembl.org/index.html	NA
CCLE (mutation status of cell lines)	http://www.broadinstitute.org/ccle/home	NA
TCGA (copy number of miRNAs)	http://www.broadinstitute.org/tcga/gistic/browseGisticByGene	NA

CONTACT FOR REAGENT AND RESOURCE SHARING

Further information and requests for resources and reagents should be directed to and will be fulfilled by the Lead Contact Piotr Sicinski (Peter_Sicinski@dfci.harvard.edu).

EXPERIMENTAL MODEL AND SUBJECT DETAILS**Animals**

For the in vivo animal studies presented here, we used NCRNU or NOG female mice (*Mus musculus*) (Taconic). 1×10^6 of CAL51, HCC1806, A549 and LoVo cells (1:1 mixture with Matrigel) were injected subcutaneously into flanks of NCRNU mice (Taconic). For patient-derived xenografts, fresh human cancer specimens were received from Brigham and Women's Hospital after signed consent was previously obtained from patients undergoing surgery, according to a research protocol approved by the Dana-Farber/Harvard Cancer Center (DF/HCC) Institutional Review Board. Fragments of the cancer specimens (2x2x2 mm) were implanted orthotopically (into the mammary fat pad, for TNBC) into female NOG mice (Taconic) or subcutaneously into the flanks of female NCRNU mice (Taconic), and were serially passaged as subcutaneous implants of tumor fragments. Dermatofibrosarcoma protuberans tumor specimens were digested with collagenase type 2 (Fisher Scientific); cells were cultured in DMEM/F12 and injected subcutaneously (1×10^6 cells) into flanks of female NCRNU mice (Taconic). All animals were held in the same animal room in Individually Ventilated Cages (IVCs) with a 12-hr day/night cycle. All procedures were carried out according to protocols approved by the Institutional Animal Care and Use Committee of the Dana-Farber Cancer Institute.

Cell Lines

Cell culture media used for each cell line is indicated in [Table S6](#).

Cell Line Authentication

Genomic DNA was extracted from 30 cell lines that were used for detailed analyses in our study, including the three cell lines used for screening, 25 breast cancer cell lines analyzed in [Figure 6](#), as well as 4 cell lines used for xenograft experiments. Short Tandem Repeat (STR) profiling was performed by the Dana-Farber Cancer Institute Molecular Biology Core. The profiles were overlaid with publicly accessible STR cell line data deposited to NCBI to verify the correct identity of these cell lines.

METHOD DETAILS

3'UTR Cloning Strategy and miR-193a PCR

The longest annotated 3'UTRs (according to TargetScan Human, Release 5.2) of the following genes: *CCNA2*, *CCNB1*, *CCNB2*, *CCND1*, *CCND2*, *CCND3*, *CCNE1*, *CCNE2*, *CDK1*, *CDK2*, *CDK4*, *CDK6*, were PCR-amplified from healthy human genomic DNA (Promega) using restriction-site flanking primers. Amplified 3'UTRs were digested with restriction enzymes and cloned into the pmirGLO basic vector (Promega). The *CDK6* 3'UTR was amplified in two separate fragments; the C-terminal fragment was digested and cloned into pmirGLO-vector using the XhoI/SbfI sites. For cloning of N-terminal *CDK6* 3'UTR fragment, an XmaI-site was introduced into pmirGLO using QuikChange XL site-directed mutagenesis kit (Agilent) followed by digestion with XmaI/XhoI. All primers, including primers used for amplification of miR-193a from healthy genomic DNA (Promega) or SW900 cells, are listed in [Table S6](#).

Screening of miRNA Library

3'UTRs-pmirGLO constructs containing 3'UTRs of cyclins or CDKs were transfected into U2OS cells. 48 hr post-transfection, cells were selected with G418 until emergence of stable clones. 20 cell clones were tested for each cyclin/CDK 3'UTR and validated by Z' tests using reverse transfection of miRNA mimics (Ambion) predicted by TargetScan to target a given 3'UTR, as well as using reverse transfection of Negative control #2 (Ambion). For each clone, the positive and negative controls were assayed in three 96-well plates each; clones were examined for expression of full-length 3'UTR, expression of firefly and renilla luciferase, and consistent reduction of the firefly/renilla ratio by a positive control. Clones that fulfilled these criteria were selected for screening.

For screening, cells were seeded in 96-well plates, 15,000 cells per well, and transfected (one miR per well, in three replicates) with human mimic pre-miR miRNA library (Ambion) containing 885 miRNAs, using Lipofectamine RNAiMAX (Invitrogen) reverse transfection. Firefly to renilla luciferase ratios were determined 28 hr post-transfection using 96-well luminometer from Turner Biosystems. Replicates of each miRNA were separated on three plates with each plate containing two replicates of negative control miRNA. Finally, firefly to renilla luciferase ratio of each miRNA was normalized to six replicates of negative control miRNA (see [Table S1](#)).

For re-screening of CAL51 and A549 cells with 20 miRNA mimics (Ambion), 3'UTRs-pmirGLO constructs were transfected into CAL51 and A549 cells, and selected as in U2OS cells. 8 cell clones were tested for each cyclin/CDK 3'UTR and for each cell line, validated and screened as in U2OS cells.

miRNAs were defined as targeting a given cyclin or CDK if they reduced the firefly to renilla luciferase ratio by at least 40%. In total, we enumerated 30 miRNAs targeting at least 5 cyclins/CDKs and 14 miRNAs targeting at least 6 cyclins/CDKs ([Figures 2A–2C](#) and [4A](#); [Table S1](#)).

In order to provide a quantitative measure of the ability of miRNAs to repress the cell cycle machinery, we also used another criterion to evaluate targeting of cyclins and CDKs. For each miRNA we calculated the average repression value across the nine screens. Sixteen miRNAs repressed all nine 3'UTRs by an average of 40% or more, while 60 miRNAs showed at least an average of 30% repression across the nine screens (please see last sheet in [Table S1](#)). The latter group of miRNAs was used for analyses shown in [Figure 3B](#); [Tables S2](#) and [S3](#).

Mutant miRNA Mimics, Mutant Cyclin/CDK 3'UTRs

Mutant miRNA mimics were synthesized by Integrated DNA Technologies by switching either a purine to a pyrimidine or a pyrimidine to a purine in position 2 of the seed sequence. Mutations of cyclin/CDK 3'UTRs were performed using Agilent XL Mutagenesis kit, by replacing six nucleotides of miRNA target sequences with an XhoI site (CTCGAG). Mutant cyclin E1 3'UTR was synthesized by Integrated DNA Technologies with flanking XhoI/XbaI restriction site sequences enabling sticky-end cloning into the pmirGLO vector. The minimal miRNA target sequence (6 nucleotides) in the cyclin E1 3'UTR was deleted (i.e., omitted during synthesis of the cyclin E1-mutant 3'UTR).

Cell growth and luciferase assays were performed as for wild-type miRNAs and wild-type cyclin/CDK 3'UTRs. Mutant 3'UTRs were transfected into U2OS cells, selected with G418 until emergence of stable clones, and validated as described above for U2OS screening clones.

Cells Expressing miRNA-Resistant Cyclin/CDKs

TNBC cell lines CAL51, BT549, MDA-MB-436, and MDA-MB-468, as well as *KRAS*-mutant lines A549 and LoVo were used for analyses. For CAL51 cells, pCMV-cyclin D1, pCMV-cyclin D3, and pCMV-cyclin E2 were transfected, followed by G418 selection until emergence of resistant clones. Clones were screened for physiological levels of ectopically expressed cyclins by RT-qPCR and used for cell expansion and cleaved caspase-3 assays as described for parental cells. BT549, MDA-MB-436 and MDA-MB-468 were transiently transfected using Lipofectamine 2000 (Invitrogen) with pCMV-cyclin D1, pCMV-cyclin D3, and pCMV-cyclin E2 (1 ng each). 24 hr post transfection, cells were transfected with miR-193a-3p using RNAiMAX (Invitrogen). For A549 and LoVo cells, pCMV-cyclin D1 and pCAG-CDK2-CDK6 (1 ng each) were transiently transfected using Lipofectamine 2000 (Invitrogen) reagent. 24 hr post transfection, cells were transfected with miR-214-5p using RNAiMAX (Invitrogen). For all experiments involving transient cyclin/CDK expression, 100,000 cells were seeded 24 hr prior DNA transfection. Cell number expansion and cleaved caspase-3 assays were performed as for parental cells.

Cell Cycle Synchronization

U2OS, CAL51 and A549 cells were synchronized at the G1/S border by double thymidine block. Thymidine was added at 2 mM concentration for 16 hr, replaced by thymidine-free medium for 10 hr, and then added for an additional 14 hr before release of cells. Cells were collected 4, 8, 12, and 15 hr after the release.

Apoptosis and Senescence Assays

Apoptosis was gauged 24 hr post-transfection using cleaved caspase-3 assays (Caspase-Glo, Promega) according to manufacturer's instructions. Senescence was monitored through staining for senescence-associated beta-galactosidase (SA- β -gal), 6 days post-transfection; 100 cells were counted in randomly chosen fields of three biological replicates.

Analyses of Cultured Cancer Cell Lines

For cell proliferation analyses shown in [Figure 4](#), 100,000 U2OS cells were transfected with the indicated miRNAs or with control miRNA (negative control #2, Ambion), each 25 nM, in triplicates. 24 hr post-transfection, cells were re-plated into 6-well plates (15,000 cells per well), and absolute cell numbers were determined every 2 days.

For assays shown in [Figures 6A and 6B](#), triple-negative breast cancer cell lines were seeded at 15,000 cells per well in 6-well plates in three replicates for each treatment. The following day, palbociclib (1 μ M, Medchem Express, NJ), dinaciclib (10 nM, ChemieTek) purvalanol A (10 μ M, Sigma-Aldrich), or DMSO were added. Drug-containing medium was changed at day 3, and the experiment was terminated at day 6. For miR-193a-3p expression, 100,000 cells were transfected with 25 nM miR-193a-3p or 25nM miRNA mimic control (Ambion, Negative control #2) in triplicates. 24 hr post-transfection, 15,000 cells were re-plated (day 0) and harvested at day 6.

ER⁺ breast cancer cell lines ([Figure 6C](#)) were plated as above, treated with palbociclib (1 μ M) or transfected with 25 nM miR-193a-3p, and harvested at day 6.

HER2⁺ breast cancer cell lines ([Figure 6D](#)) were plated as above and treated with palbociclib or lapatinib (Haoyuan Chemexpress) alone (each 1 μ M), a combination of palbociclib plus lapatinib, or transfected with 25 nM miR-193a-3p. Cell numbers were determined after 6 days.

The data were presented as follows. We determined the absolute number of cells at day 6 in drug-treated and DMSO-treated cells, as well as in miR-193a-3p-transfected and control miRNA-transfected cells. By dividing the cell number in experimentally treated (drugs, miR-193a-3p) versus control-treated cells, we calculated the reduction in cell number expansion (in [Figure 6](#)).

CCLE Cell Line Screen

Cells were transfected with mirVana miRNA mimic miR-193a-3p, miR-195-5p, miR-214-5p, miR-890, or with control miRNA (Negative control #2, Ambion), all at 25 nM, and either re-plated 24 hr post-transfection or left unperturbed, depending on cell growth rates. All cell lines were plated at a fixed cell number prior to transfection and, if re-plated, 24 hr post-transfection. For cell number expansion assays, experiments were terminated 6 days post miRNA transfection and cells were counted using a Beckman Coulter Z series dual threshold analyzer. All cell lines used in CCLE screen were assayed using a minimum of three biological replicates.

The data were presented as follows. For cell number expansion ([Figure 5A](#)), we determined cell numbers at day 6 post-transfection in cells transfected with a given cell cycle-targeting miRNA, as well as in cells transfected with a negative control. The fold-reduction of cell expansion was then calculated by dividing cell number in cells transfected with control miRNA by cell number in cells transfected with cell cycle-targeting miRNA.

For induction of apoptosis ([Figure 5B](#)), cleaved caspase-3 assays (Casp-Glo, Promega) were performed 24 hr after transfection. The values observed in cells expressing a given cell cycle-targeting miRNA were divided by values seen in cells transfected with control miRNA, and fold-increase was calculated and presented.

For induction of senescence, cells were stained for senescence-associated beta-galactosidase (SA- β -gal), according to manufacturer instructions (Biovision), at 6 days post-transfection. 100 cells in 3 different fields of view were counted and fold-increase was calculated and presented in [Figure S5A](#).

Non-transformed cell lines were transfected, treated and assayed as for CCLE cell line screen.

Western Blotting

For western blot analysis, the following dilutions were used for the primary antibodies (see also [Key Resources Table](#)): cyclin D1 (1:5000), cyclin D3 (1:1000), cyclin E1 (1:1000), cyclin E2 (1:1000), cyclin A2 (1:5000), cyclin B1 (1:500), CDK1 (1:500), CDK2 (1:1000), CDK4 (1:1000), CDK6 (1:500), GAPDH (1:2500), and beta-actin (1:10000).

miRNA Transfection and miRNA qPCR

Pre-miR miRNA mimic (Ambion) or mirVana miRNA mimic (Ambion) were transfected using Lipofectamine RNAiMAX (Invitrogen; standard forward transfection) with a final miRNA concentration of 25 nM. Total RNA and miRNA were isolated using mirVana miRNA extraction kit (Ambion). miRNA stem loop RT-qPCR was performed using mirVana miRNA assays (Ambion) according to manufacturer's instructions, normalized to U6-sRNA (as internal small RNA control).

RT-qPCR Analysis

RT-qPCR analysis was performed using the delta-delta-Ct-method with GAPDH as a reference gene. Reverse transcription was carried out using high-capacity RNA-to-cDNA kit with random primers, and qPCR with SYBR green master mix (Applied Biosystems). Analysis was performed using the Applied Biosystems 7300 Real-Time PCR System.

Flow Cytometry Analysis

For analyses of cell lines, cells were pulsed with BrdU for 1 hr, stained with anti-BrdU antibody conjugated to fluorescein isothiocyanate (FITC) (BrdU flow kit, BD Biosciences), according to manufacturer's instructions, and with propidium iodide, and analyzed by FACS. Data acquisition was performed on LSRII, Fortessa, or Facscan, and analyzed using Cell Quest and FACSDiva software (BD Biosciences).

Formulation of miRNAs into Nanoparticles

In vivo ready miR-193a-3p, miR-214-5p and mirVana miRNA mimic Negative control #1 were purchased from Ambion and formulated into C12-200 nanoparticles. miRNA mimics were encapsulated in C12-200 lipid nanoparticles using a similar method as previously described for siRNA encapsulation (Love et al., 2010). Briefly, nanoparticles were synthesized by mixing together an ethanol phase containing the ionizable lipidoid C12-200, 1,2-distearoyl-sn-glycero-3-phosphocholine (DSPC), cholesterol, and 1,2-dimyristoyl-sn-glycero-3-phosphoethanolamine-N [methoxy(polyethylene glycol)-2000] (ammonium salt) (C14-PEG 2000) at a 50:10:38.5:1.5 molar ratio with an aqueous phase containing mimic in citrate buffer (pH 3.0) at a 5:1 C12-200:mimic weight ratio. Ethanol and aqueous phases were mixed at a 1:3 volume ratio in a microfluidic chip device.

C12-200 lipid nanoparticles had an average mimic encapsulation efficiency of 70-80%, determined by a modified Quant-iT Ribo-green RNA assay. The nanoparticles had an average intensity diameter of approximately 130 nm and polydispersity (PDI) between 0.1 and 0.2. Size and PDI were measured using dynamic light scattering (ZetaPALS, Brookhaven Instruments).

Nanoparticle-Mediated Delivery of miRNAs

Mice with palpable tumors were injected via the tail vein with C12-200-miRNA-formulated nanoparticles (1.5 mg/kg of body weight) in a total volume of 200 μ L, every other day, until tumors in control group reached 20 mm in any direction. The dose and dosing schema were selected based on prior published work from the Anderson laboratory (Love et al., 2010).

Tumors were measured by caliper every other day. Four hr before sacrifice, mice were injected intraperitoneally with 2 mg of BrdU. A total of 179 mice were used for in vivo analyses: 93 tumor-bearing mice were treated with nanoparticles containing miR-193a-3p or miR-214-5p and 86 mice with control nanoparticles.

Analyses of miRNA Uptake In Vivo

Organs and tumors were dissected from animals treated with nanoparticles containing control miRNA (Ambion, Negative control #1), miR-193a-3p, or miR-214-5p, at the end of treatment period (20-34 days). RNA was extracted using mirVana miRNA extraction kit (Ambion). miRNA stem loop RT-qPCR was performed using mirVana miRNA assays (Ambion) according to manufacturer's instruction, with U6-snrRNA as internal small RNA control. In each organ or tumor, we determined the delta-Ct values relative to U6-snrRNA. Delta-Ct values were then transformed into percentage expression relative to U6-snrRNA. For example, an organ displaying a Ct value of 20 for miR-193a-3p and a Ct value of 18 for U6-snrRNA has the delta-Ct value of 2, which is transformed into a percentage expression relative to U6-snrRNA of 25 %.

Analyses of Liver Cytokine mRNA Levels

RNA was extracted from livers of mice treated with nanoparticles containing control miRNA (Ambion, Negative control #1) or miR-214-5p, and transcript levels for the indicated cytokines (Figure S8H) levels were measured using the delta-delta-Ct method with GAPDH as a reference gene.

Embedding, Sectioning and Staining

Tumors were fixed in 10% neutral-buffered formalin for 36 hr, followed by transfer to 70% ethanol. The Rodent Histopathology Core at Harvard Medical School carried out paraffin embedding, sectioning, and staining with hematoxylin and eosin (H&E). The Brigham and Women's Hospital Specialized Histopathology Core performed staining for BrdU, Ki67 and cleaved caspase-3. Tumors and normal organs from a minimum of three mice were analyzed per treatment group. Each tumor and normal organ was analyzed for histology (H&E staining) as well as for BrdU, Ki67, and cleaved caspase-3 staining. Each mouse received a minimum of ten miRNA nanoparticle doses (1.5mg/kg per dose).

Hematological Analyses

Mice were treated with nanoparticles containing control miRNA (Negative control #1, Ambion), miR-193a-3p, or miR-214-5p for 17 days. Six mice per group were analyzed together with an untreated group (also n=6). Blood was collected by cardiac puncture, divided into samples for whole blood count analysis (using BD Microtainer tubes with K2-EDTA) and for blood serum analysis (using BD Microtainer serum separator tubes), and analyzed by Department of Laboratory Medicine of the Boston Children's Hospital.

Computational Analyses

TargetScan Context+ Score Analysis

TargetScan Human Release 6.2 was used. All miRNAs used in our screen were analyzed for predicted targeting of the cyclin D1 3'UTR using predicted context+ scores.

Target Prediction by Multiple Programs

Besides TargetScan, prediction programs miRanda, miRDB, miRWalk, and TargetSpy were used to predict targeting of cyclin D1 3'UTR (Figures S3D and S3E). In Figure S3F, these algorithms were used to predict targeting of the indicated seven cyclin/CDK 3'UTRs. CDK1 and CDK4 were excluded from the analyses due to their short 3'UTRs (<500 nt). For combined prediction of TargetScan and miRanda, miRNAs that were predicted to target the indicated gene by both TargetScan and miRanda algorithms were considered as targeting miRNAs. The screen ratios of “predicted targeting” versus “predicted non-targeting” groups were compared using t-test; the p values are shown in Figure S3F.

Hierarchical Clustering

Hierarchical clustering was applied to both cyclins/CDKs and miRNAs. The distance between each gene/miRNA was computed as Euclidean distance and the linkage option was complete linkage.

Permutation Analysis

For each permuted screening assay, screening scores were randomly permuted and assigned to miRNAs. We performed 10,000 random permutations and calculated the numbers of miRNAs targeting multiple cyclins/CDKs in each permutation. For example, we observed in total 7 miRNAs targeting 7 cyclins/CDKs in all 10,000 permutations, so we estimated that 0.0007 (7/10,000) miRNAs would target 7 cyclins/CDKs by chance. Figure 2B shows the mean and SE from 10,000 permutations. The permutation p value was estimated as the proportion of permutations in which the number of miRNAs targeting multiple cyclins/CDKs is equal to or greater than the observed number.

In the same way, we calculated the number of miRNAs exclusively targeting each cyclin or CDK from 10,000 permutations (Figure S3I). The permutation p value was estimated as the proportion of permutations in which the number of miRNAs exclusively targeting this gene is equal to or greater than the observed number.

Pairwise Comparison

For each pair of cyclins/CDKs, we determined the targeting miRNAs by a threshold of 40% repression. Then, a contingency table was constructed based on the threshold, and Fisher's exact test was applied to the table to obtain the p value.

Guilt-by-Association Analysis

For each miRNA used in the screens, the Spearman correlation between the levels of this miRNA and the levels of all transcripts were computed across 4,807 human tumor samples representing 18 tumor types. Then, the genes were ranked based on the correlations. The GSEA analysis was used to calculate the p values of the ranked gene list's enrichment in KEGG pathways. For enrichment among highly correlated genes, we marked the corresponding miRNA-pathway blocks in red, and for enrichment among highly anti-correlated genes, we marked the blocks in blue. p values were calculated by hypergeometric test. Hypergeometric test uses the hypergeometric distribution to calculate the statistical significance of having drawn a specific *k* successes (out of *n* total draws) from the aforementioned population. The test is used to identify which sub-populations are over- or under-represented in a sample.

miRNA Anti-Correlation with Cyclin/CDK mRNA

To integrate anti-correlations between miRNA and cyclins/CDKs in 18 cancer types, a MDS (multiple-dimensional scaling) method was used. At first, for each cancer type, expression data for miRNAs and cyclins/CDKs was merged. Then the distance between a miRNA and a gene was calculated for MDS as $1 + \text{correlation}$, so stronger negative correlation was associated with closer distance. The miRNAs were ranked based on weighted sum of the distances to a cyclin/CDK gene across all cancer types, where the weights were the ratio between a cancer type's sample number and the total sample number. The expression data (RNAseqV2 data) were obtained from TCGA. For analysis, we used miRNAs, which showed mean repression level greater than 30% across the nine screens. Cancer types included in this study were BLCA, BRCA, CESC, COAD, HNSC, KICH, KIRC, KIRP, LIHC, LUAD, LUSC, PAAD, PRAD, READ, SARC, SKCM, THCA, and UCEC. CESC, cervical squamous cell carcinoma and endocervical adenocarcinoma; COAD, colon adenocarcinoma; PAAD, pancreatic adenocarcinoma; READ, rectal adenocarcinoma; SARC, sarcoma; SKCM, skin cutaneous melanoma. For other tumor types, please see legend to Figure 3B.

Analysis of miRNA Downregulation in Tumors

miRNAs expression levels were compared between normal and tumor samples for 14 cancer types. Comparison was based on Wilcoxon test. For analysis, we used miRNAs, which showed mean repression level greater than 30% across the nine screens (Table S1).

KEGG Pathway Analysis

For analysis, we used miRNAs, which showed mean repression level greater than 30% across the nine screens (Table S1). Among these, we were able to predict targets using TargetScan for 47 miRNAs corresponding to 40 families. We also used TargetScan to predict targets of 200 randomly selected other miRNAs. The targets of each miRNA were then analyzed for KEGG pathways enrichment. For each pathway, the numbers of miRNAs whose targets were enriched in this pathway were counted for 40 cell cycle-targeting miRNAs and for 200 other miRNAs, and the ratios were compared using Fisher's exact test. The proliferative pathways are marked in yellow in Table S2.

TCGA Copy Number Analysis of miRNAs

For miRNA copy number analysis, miRNAs were analyzed by GISTIC, available through the Broad Institute TCGA copy number portal. Data was filtered for Q-values less than 0.25. To search for cell cycle-targeting miRNAs that are commonly deleted in human

cancers, for each miRNA we plotted the distance between the locus encoding this miRNA to the nearest deletion peak versus the effect of this miRNA across the nine cyclin/CDK-luciferase screens (Figure 3C).

Gene Expression Array, RNA Sequencing

A fresh dermatofibrosarcoma tumor was homogenized and total RNA isolated using Ambion mirVana miRNA isolation kit (without enrichment for small RNAs). Total RNA was hybridized on Affymetrix GeneChip Human Genome U133 Plus 2.0 Array, in three technical replicates. Raw intensity files were analyzed using the R packages Affy and limma, and the Bioconductor software. For RNA sequencing, 100,000 SW900 cells were transfected with 25 nM of miR-193a-3p mimic or 25 nM of negative miRNA control (Negative control #2, Ambion). 48 hr post-transfection, total RNA was isolated using Ambion mirVana miRNA isolation kit without enrichment for small RNAs. RNA was sequenced on an Illumina HiSeq2000 at the Center for Cancer Computational Biology at the Dana-Farber Cancer Institute.

Cell Line Mutational Status Correlation

Cell lines' mutation statuses were obtained from CCLE. For each gene's mutation, cell lines were divided into wild-type and mutant groups. The response to miRNA expression (fold-decrease in cell number expansion) was compared between the two groups using Wilcoxon test.

Derivation of an Expression-Based Algorithm

An elastic net regression model was used to predict cell lines response to miRNA expression (see: Table S5 for prediction results). The model's predictors were cell lines expression data (scaled to make each gene have mean expression level 0 and SD 1) and the response was cell lines response to miRNA (day 6 cell number in control group / cell number in treated group). For a total of 117 cell lines, 90 cell lines were randomly picked as a training set. In the training set, correlation coefficients between each gene's expression level and cell lines' response to miRNA expression were calculated and only the genes with absolute correlation coefficient greater than 0.1 were retained for regression. We ran 200 times bootstrap and applied elastic net model to the resampled data. Each time only a small fraction of genes had significant coefficients and were retained in the model. Then we counted the times a gene was retained and choose the most frequently (more than 35%) retained genes as features. Finally, a 10-fold cross-validation was applied to choose regularization parameters and the final model was fitted using the whole dataset before predicting new samples.

QUANTIFICATION AND STATISTICAL ANALYSIS

Quantification of IHC Staining

Quantification of IHC staining was performed in a blinded fashion for all experiments. The fraction of cells stained for Ki67, BrdU, or cleaved caspase-3 was determined by manual counting of histological sections stained with the respective antibodies.

Statistical Analysis

For analysis of gene expression from primary dermatofibrosarcoma tumor, raw intensity files were analyzed using the R packages Affy and limma, as well as the software Bioconductor. For RNA sequencing data, reads were mapped to hg19 reference genome and gene expression was quantified by RSEM v1.2.25. Differential expression analysis was conducted by EBSeq version 1.10.0. Differentially expressed genes were defined as having PPDE (posterior probability of differential expression) equals to 1. Gene Ontology analysis was conducted by BiNGO version 3.0.3.

All computational analyses have been described in the subsection "Computational Analyses" of the section "Method Details".

Statistical significance was defined as a p value of less than 0.05 using the appropriate statistical test method. For each experiment, the statistical test is indicated in the figure legend. Error bars are represented as either mean $\pm$ SD or mean $\pm$ SEM, and this is indicated in the figure legend.

DATA AND SOFTWARE AVAILABILITY

Deposited Data

The accession numbers for the gene expression array and RNA sequencing data reported in this paper are GSE77228, GSE77229 and GSE77230.

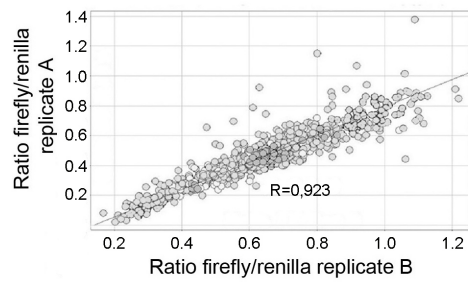
Supplemental Information

Cell-Cycle-Targeting MicroRNAs as Therapeutic

Tools against Refractory Cancers

Per Hydbring, Yinan Wang, Anne Fassl, Xiaoting Li, Veronica Matia, Tobias Otto, Yoon Jong Choi, Katharine E. Sweeney, Jan M. Suski, Hao Yin, Roman L. Bogorad, Shom Goel, Haluk Yuzugullu, Kevin J. Kauffman, Junghoon Yang, Chong Jin, Yingxiang Li, Davide Floris, Richard Swanson, Kimmie Ng, Ewa Sicinska, Lars Anders, Jean J. Zhao, Kornelia Polyak, Daniel G. Anderson, Cheng Li, and Piotr Sicinski

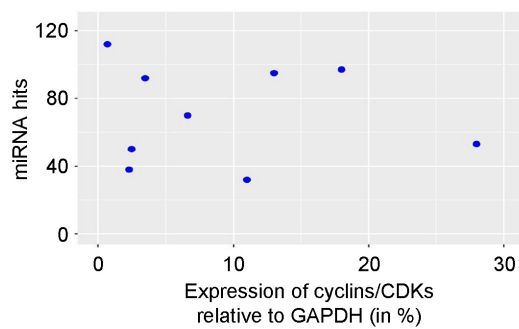
A



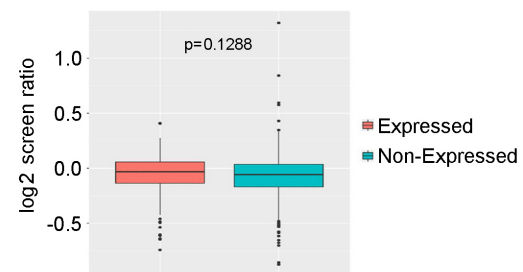
B

	Expression relative to GAPDH (in %)	Number of hits
Cyclin D1	13	95
Cyclin D2	28	53
Cyclin D3	2	38
Cyclin E1	3	50
Cyclin E2	1	112
CDK1	11	32
CDK2	4	92
CDK4	18	97
CDK6	7	70

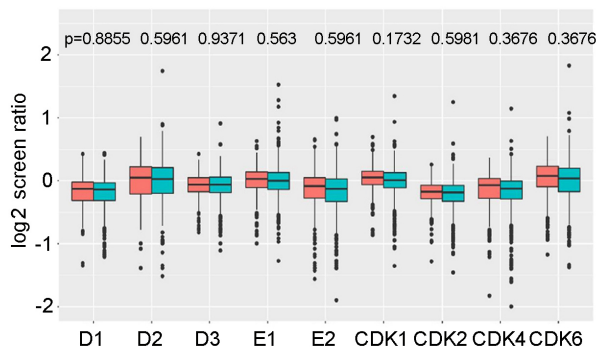
C



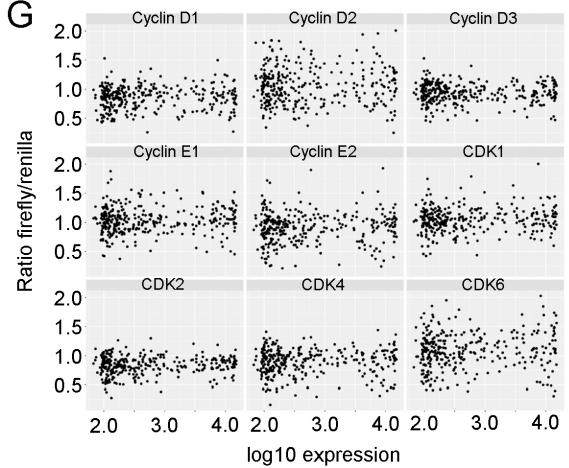
D



E



G



F

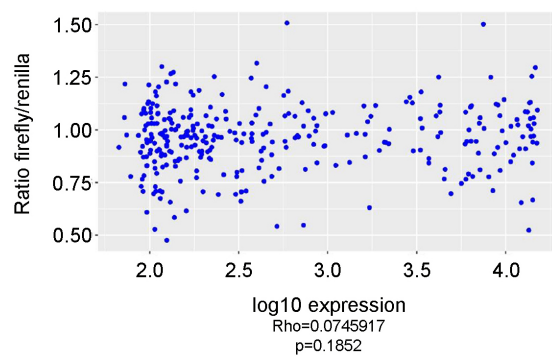


Figure S1, Related to Figure 1. Screening for miRNAs Regulating Cyclins and CDKs and Biocomputational Analyses of Screening Results

(A) Correlation plot of firefly to renilla luciferase ratios between replicates A and B of the cyclin D1 3'UTR screen. Like all our screens, the cyclin D1 screen was performed in three replicates (A, B and C).

(B and C) These panels search for a correlation between expression levels of endogenous cyclins and CDKs in U2OS cells versus the numbers of miRNAs that were found to target each cyclin or CDK in our U2OS-based screens. (B) A table and (C) a scatterplot showing endogenous expression levels of the indicated cyclins and CDKs in U2OS cells versus the numbers of miRNAs that targeted each cyclin and CDK, observed in our 3'UTR screens (Table S1). The levels of endogenous cyclin and CDK transcripts in U2OS cells are shown relative to expression levels of an endogenous housekeeping gene GAPDH in these cells. Thus, 1% of GAPDH expression corresponds to 1/100 of GAPDH levels. In (C) each dot corresponds to a different cyclin or CDK. Pearson's $r = -0.0886$, p value = 0.8207.

(D) This panel tests whether miRNAs that are expressed in U2OS cells gave different screening results than miRNAs that are not expressed in U2OS cells, across the nine screens. Shown are boxplots of $\log_2$ values of the average firefly/renilla luciferase ratios across the nine cyclin/CDK screens. The comparison is between miRNAs that are expressed in U2OS cells versus miRNAs that are not expressed. $p=0.1288$, using unpaired t-test. The box boundaries and the middle thick line represent the 25th percentile, the 75th percentile and the median of the data points. The upper and lower lines indicate the largest or the smallest value within 1.5 the interquartile range from the 75th or the 25th percentile. The values outside these two lines are regarded as outliers, indicated by individual circles.

(E) This panel tests – individually for each of the nine screens - whether miRNAs that are expressed in U2OS cells gave different screening results than miRNAs that are not expressed in U2OS. Shown are boxplots of $\log_2$ values of the average firefly/renilla luciferase ratios observed in each cyclin or CDK screen. The comparison is between miRNAs that are expressed in U2OS cells versus miRNAs that are not expressed ($p=0.8855$ for cyclin D1, 0.5961 for cyclin D2, 0.9371 for cyclin D3, 0.5630 for cyclin E1, 0.5961 for cyclin E2, 0.1732 for CDK1, 0.5981 for CDK2, 0.3676 for CDK4, and 0.3676 for CDK6, using unpaired t-test). For explanation of box boundaries, see (D). D1, cyclin D1; D2, cyclin D2; D3, cyclin D3; E1, cyclin E1; E2, cyclin E2.

(F) This panel analyzes only miRNAs that are expressed in U2OS cells, and searches for a correlation between the expression levels of a given miRNA in U2OS cells versus the results for this miRNA across the nine screens. Each dot corresponds to a different miRNA. Horizontal axis depicts $\log_{10}$ values of the expression levels of the endogenous miRNAs in U2OS cells, vertical axis: average firefly/renilla luciferase ratios across the nine screens when this miRNA was transfected into U2OS cells. Spearman's $\rho=0.0746$, p value=0.1852.

(G) This panel analyzes only miRNAs that are expressed in U2OS cells, and searches for a correlation between the expression levels of a given miRNA in U2OS cells versus the results for this miRNA in each of the nine screens. Each dot corresponds to a different miRNA. Horizontal axes depict $\log_{10}$ values of endogenous miRNA expression levels in U2OS cells, vertical axes: firefly/renilla luciferase ratios in the indicated screen when this miRNA was transfected into U2OS cells. Spearman's ρ from left to right and top to bottom: 0.0509, 0.0090, -0.0375, 0.0867, 0.0866, 0.0638, 0.0578, 0.0471, 0.0966; p value: 0.3667, 0.8726, 0.5063, 0.1236, 0.1237, 0.2578, 0.3048, 0.4032, 0.0860.

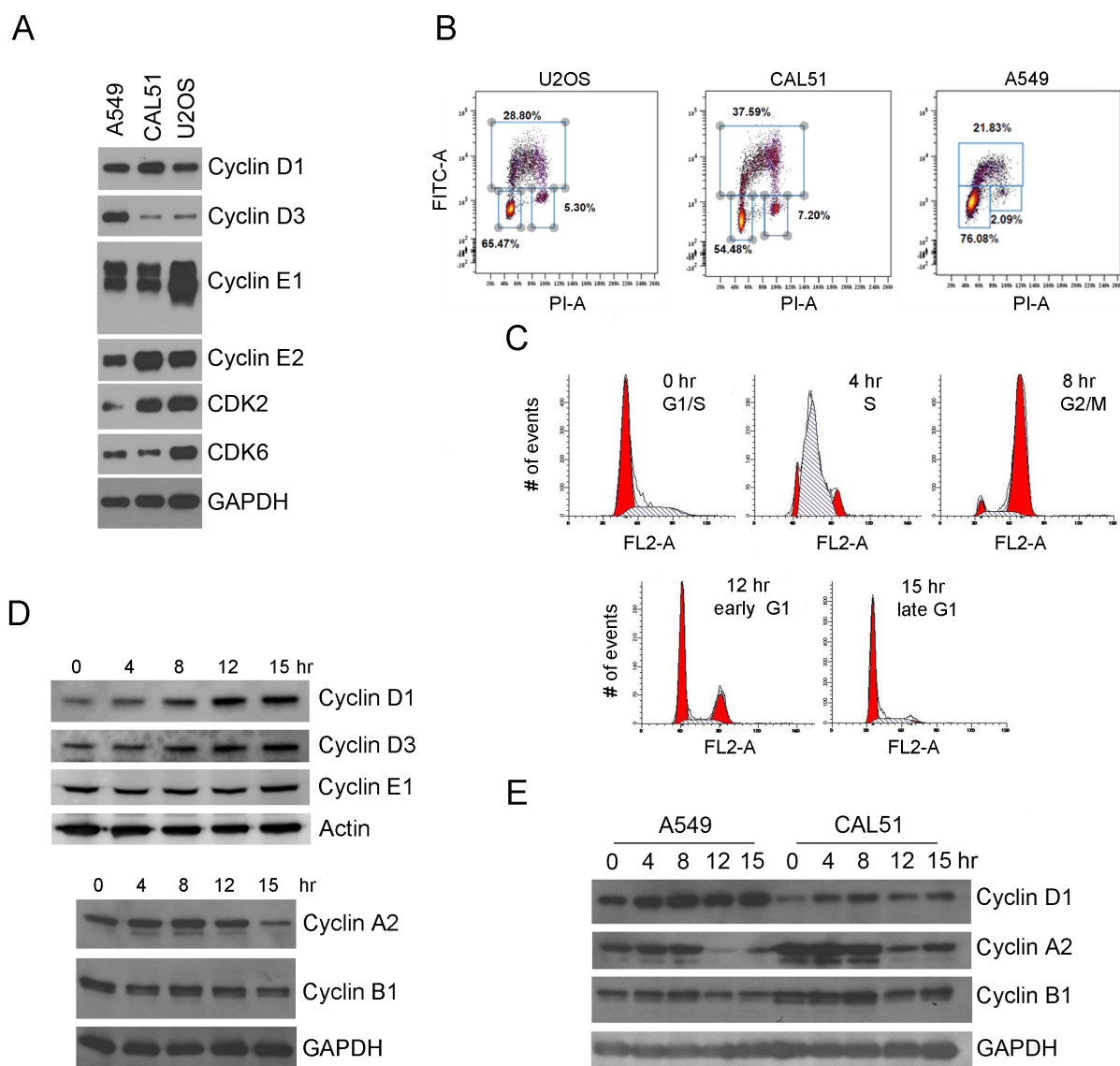


Figure S2, Related to Figure 1. Cell Cycle Analyses of Cell Lines Used for Screening

(A) Immunoblot analysis of cyclins and CDK levels in A549, CAL51 and U2OS cells.

(B) Cells were pulsed with bromodeoxyuridine (BrdU), stained with anti-BrdU antibodies and with propidium iodide and analyzed by FACS. Shown are percentages of cells in the indicated cell cycle phases.

(C) U2OS cells were synchronized at the G1/S boundary by double-thymidine block, and released. At the indicated time-points, cells were stained with propidium iodide and analyzed by FACS to gauge progression through the cell cycle (the majority of cells in S phase after 4 hr, in G2/M after 8 hr and in G1 after 12 and 15 hr).

(D) Immunoblot analysis of cells from (C). Cells collected at different time-points after release were examined for expression of the indicated cell cycle proteins.

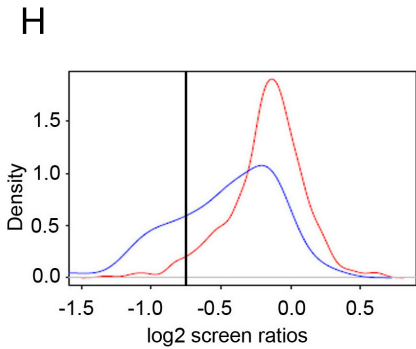
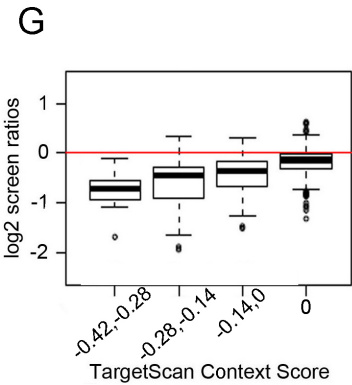
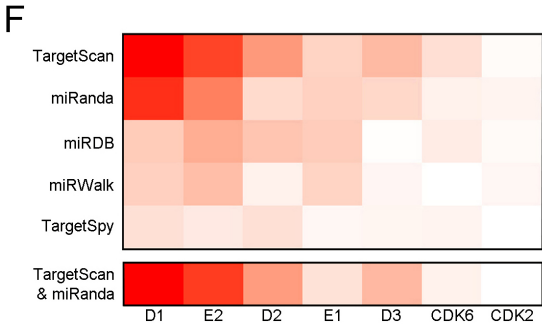
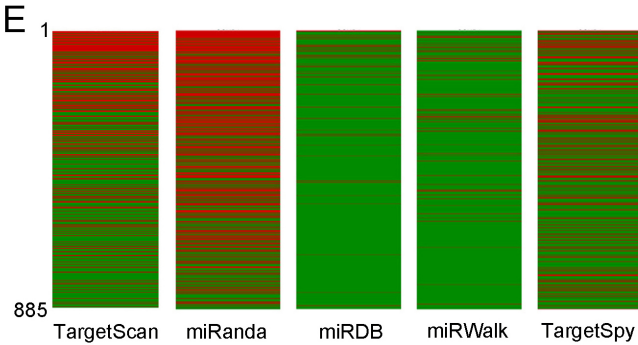
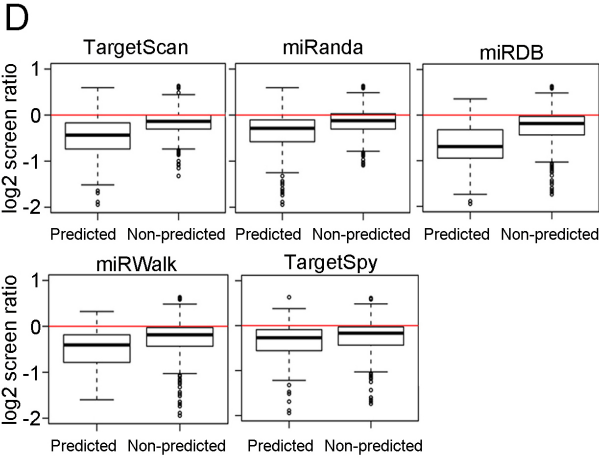
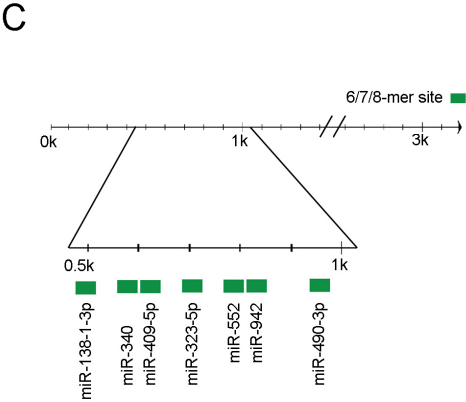
(E) Similar analysis as in (D) using synchronized A549 and CAL51 cells.

A

	U2OS vs. CAL51	U2OS vs. A549	A549 vs. CAL51
Cyclin D1	85%	80%	75%
Cyclin D2	75%	80%	75%
Cyclin D3	90%	75%	65%
Cyclin E1	95%	90%	90%
Cyclin E2	70%	85%	75%
CDK1	90%	90%	90%
CDK2	65%	45%	70%
CDK4	55%	60%	55%
CDK6	90%	90%	85%
Mean	79.4%	78.3%	75.6%

B

	U2OS vs. CAL51	U2OS vs. A549	A549 vs. CAL51
Cyclin D1	80%	80%	75%
Cyclin D2	60%	80%	75%
Cyclin D3	75%	95%	85%
Cyclin E1	70%	90%	75%
Cyclin E2	65%	65%	65%
CDK1	75%	75%	100%
CDK2	80%	60%	70%
CDK4	70%	80%	70%
CDK6	60%	85%	70%
Mean	70.6%	78.9%	76.1%



I

	Unique hits screen	Permutation	p value
Cyclin D1	44	52.4	0.9685
Cyclin D2	21	28.0	0.9805
Cyclin D3	4	19.7	1.0000
Cyclin E1	10	25.3	1.0000
Cyclin E2	43	60.0	0.9999
CDK1	2	15.3	1.0000
CDK2	29	47.1	1.0000
CDK4	9	50.0	1.0000
CDK6	17	34.8	1.0000

Figure S3, Related to Figure 1. Analyses of Screening Results

(A) Comparison of the results of the original U2OS-based screen versus the results in CAL51 and A549 cells. U2OS, CAL51 and A549 cells were engineered to stably express firefly luciferase upstream of 3'UTRs for each of the nine cyclins or CDKs (total of $3 \times 9 = 27$ reporter cell lines). We then used these reporter cell lines to validate the nine U2OS-based original screens. For instance, to validate U2OS-based cyclin D1 screen, selected 20 miRNAs were transfected into CAL51-based and A549-based reporter cell lines containing cyclin D1 3'UTR, and luciferase activity was measured. As it was the case in the original U2OS-based screen, repression of firefly luciferase (i.e. reduction of firefly to renilla ratio) by 40% or more was considered as a "hit", and by less than 40% as a "non-hit". We considered the result "reproduced" if a given miRNA scored in both cell lines as a "hit", or it scored in both cell lines as a "non-hit". Figure 1B in the main text illustrates reproducibility of "hits" and "non-hits" for each miRNA. Here, the overall reproducibility for 20 miRNAs tested against cyclin D1 3'UTR is shown between U2OS vs. CAL51 (85%), U2OS vs. A549 (80%), and A549 vs. CAL51 (75%). In the similar way we validated the remaining screens. Reproducibility for cyclin D2, D3, E1, E2, CDK1, CDK2, CDK4 and CDK6 screens was analyzed as for cyclin D1, and the data are presented for each screen and for each pair of cell lines being compared, along with mean reproducibility for each pair of cell lines across all nine screens. For each miRNA and cyclin/CDK, $n=3$.

(B) Comparison of the results of the original U2OS-based screen versus the results in CAL51 and A549 cells, as above. The data was analyzed in a different way than in (A). Here, we considered the result as "reproduced" if the repression values observed in the two cell lines were within 25% from each other. For instance, we considered a result reproduced if luciferase/renilla ratios in U2OS and CAL51 cells were, respectively: 0.1 and 0.2, or 0.4 and 0.5, or 0.7 and 0.8. We considered not reproduced, if the respective ratios were: 0.1 and 0.36, or 0.2 and 0.46, or 0.7 and 0.96. Shown is reproducibility for each screen and for each pair of cell lines being compared, along with mean reproducibility for each pair of cell lines across all nine screens. For each miRNA and cyclin/CDK, $n=3$.

(C) A Schematic representation of cyclin D1 3'UTR. An enlarged panel shows examples of predicted miRNA targeting based on manual mapping of 6/7/8-mer seed sites.

(D) Boxplots of log2 values of the repression (firefly to renilla luciferase ratios) seen in our cyclin D1 3'UTR screen (vertical axis) stratified by prediction results of the indicated softwares (predicted to target cyclin D1 3'UTR vs. non-predicted). Red line represents no repression. The box boundaries and the middle thick line represent the 25th percentile, the 75th percentile and the median of the data points. The upper and lower lines indicate the largest or the smallest value within 1.5 the interquartile range from the 75th or the 25th percentile. The values outside these two lines are regarded as outliers, indicated by individual circles. For each miRNA, $n=3$.

(E) Overlaying the results of our cyclin D1 3'UTR screen with predictions of TargetScan, miRanda, miRDB, miRWalk and TargetSpy. Each horizontal row corresponds to a different miRNA. miRNAs were arranged based on the results of our screen, with miRNA most potently repressing cyclin D1 3'UTR at the top. miRNAs predicted by the indicated software to target cyclin D1 3'UTR are marked in red, miRNAs predicted not to target this 3'UTR are in green. For each miRNA, $n=3$.

(F) A heatmap of comparisons of the repression (firefly/renilla luciferase ratios) observed in the indicated cyclin or CDK screens, between targeting miRNAs versus non-targeting miRNAs predicted by the indicated softwares. Color scale depicts $-\log_{10}$ unpaired t-test p value, with most significant difference in deep red, which indicates a good prediction. X- and y-axes were ordered based on ranks of overall prediction results (mean $-\log_{10}$ p value). D1, cyclin D1; D2, cyclin D2; D3, cyclin D3; E1, cyclin E1; E2, cyclin E2.

(G) TargetScan analysis of cyclin D1 3'UTR screen based on context+ score categorization. miRNAs were classified into four categories, based on TargetScan context score (horizontal axis). Decreasing (negative) TargetScan scores indicate higher probability of targeting; three transcripts with context+ score < -0.42 are not shown, since this number is too low for statistical analysis. The category with TargetScan "0" score (first on the right) is predicted not to target cyclin D1 3'UTR. The vertical axis denotes log2 values of the repression of cyclin D1 3'UTR observed in our screen, with the lowest (negative) values representing the strongest repression. Red line corresponds to no repression. The distribution of values is shown as boxplots. For explanation of box boundaries, see (D). For each miRNA, $n=3$.

(H) The distribution of cyclin D1 3'UTR screen values of "predicted-targeting" and "predicted non-targeting" miRNA groups, predicted by TargetScan. Horizontal axis denotes log2 values of the repression observed in our screen (strongest repression on the left). A thick black vertical line marks the cutoff of 40% repression, with miRNAs showing $>40\%$ repression located on the left of the line. Blue color depicts miRNAs predicted by TargetScan to target cyclin D1, red color miRNAs predicted not to target cyclin D1. For each miRNA, $n=3$.

(I) Enrichment analysis of miRNAs uniquely targeting individual cyclins or CDKs. Shown is the number of miRNAs observed in our screens to uniquely target the indicated gene (Unique hits screen), the expected number of miRNAs based on permutation analysis (Permutation), and permutation p values.

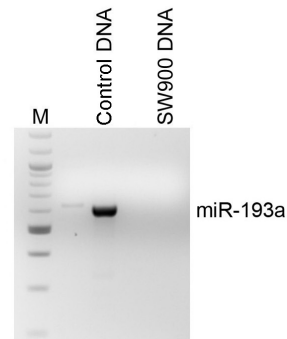
Table S1, Related to Figure 1. Results of the Screens for miRNAs Targeting Cyclins and CDKs. Provided as an Excel file.

Table S2, Related to Figure 2. Comparison of KEGG Pathway Enrichment for Predicted Targets of Cell Cycle-Targeting miRNAs. Provided as an Excel file.

A

Cell Line	miR-193a log2
SW900	-4.167
YD15	-1.601
SCC4	-1.464
SNU119	-1.382
KURAMOCHI	-1.314
ABC1	-1.146
HCC1806	-1.062
MFE280	-1.058
F36P	-1.042
JHOM2B	-1.038

B



C

KEGG term	Number of transcripts	p value	Genes
Cell cycle	28	4.46E-24	E2F1, YWHAZ, PKMYT1, TTK, CHEK1, PTTG1, CDC45, BUB1, CCNA2, CDC7, CDC6, CDK1, RBL1, CDK6, ESPL1, CDC20, MCM2, MCM3, CDK2, CDC25A, MCM5, CCNB1, CCND1, CCNB2, MAD2L1, PLK1, PCNA, BUB1B
DNA replication	12	5.50E-12	PRIM1, RFC3, POLE2, POLE3, LIG1, PCNA, POLA2, MCM2, MCM3, RNASEH2A, FEN1, MCM5
Oocyte meiosis	15	1.91E-09	CDK1, YWHAZ, SGOL1, PKMYT1, AURKA, CDC20, ESPL1, PTTG1, CDK2, CCNB1, CCNB2, MAD2L1, PLK1, BUB1, FBXO5
Progesterone-mediated oocyte maturation	10	9.70E-06	CCNB1, CDK1, MAD2L1, CCNB2, PLK1, BUB1, PKMYT1, CCNA2, CDC25A, CDK2
Base excision repair	7	1.88E-05	POLE2, POLE3, UNG, NEIL3, LIG1, PCNA, FEN1
Nucleotide excision repair	5	5.69E-03	RFC3, POLE2, POLE3, LIG1, PCNA
Small cell lung cancer	6	1.21E-02	E2F1, CCND1, CDK6, LAMC2, BIRC3, CDK2

Figure S4, Related to Figure 3. Analyses of miR-193a-3p

(A) Log2-copy number changes from Cancer Cell Line Encyclopedia (CCLE). Shown are top ten cell lines harboring deletions of the miR-193a locus.

(B) PCR-based detection of the miR-193a locus using human genomic DNA (Control) or DNA from SW900 cell line as templates. M, size marker. 100 bp DNA ladder (New England Biolabs).

(C) Significantly enriched KEGG terms for transcripts that were downregulated upon re-introduction of miR-193a-3p into SW900 cells. Gene symbols for the downregulated transcripts belonging to each term are also shown. p values were obtained by hypergeometric test.

Table S3, Related to Figure 3. Identification of miRNAs Anti-Correlated with Cyclins and CDKs in Expression. Provided as an Excel file.

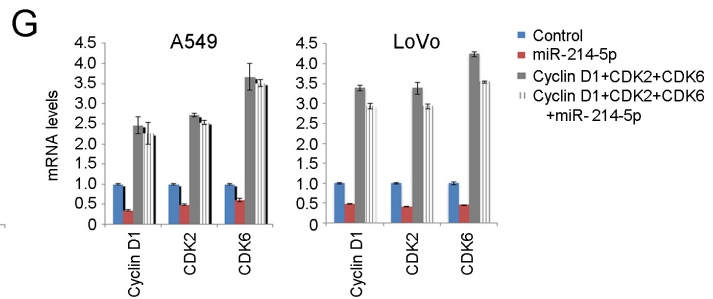
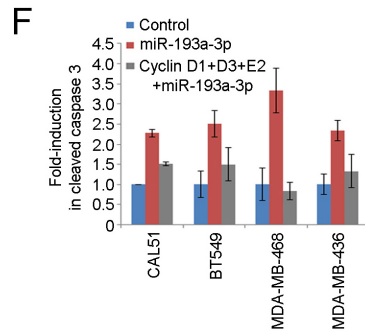
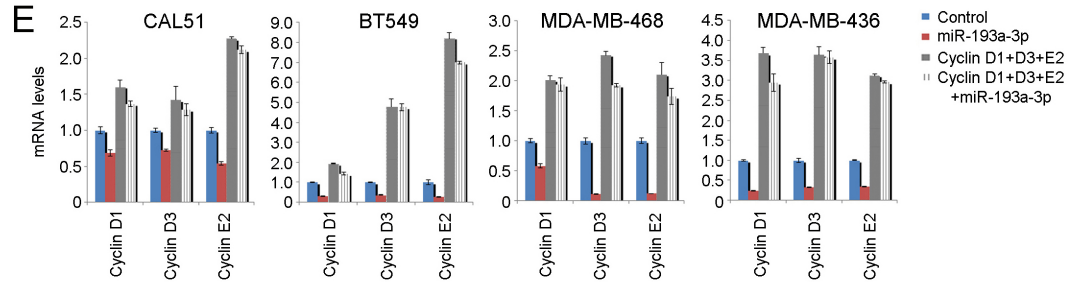
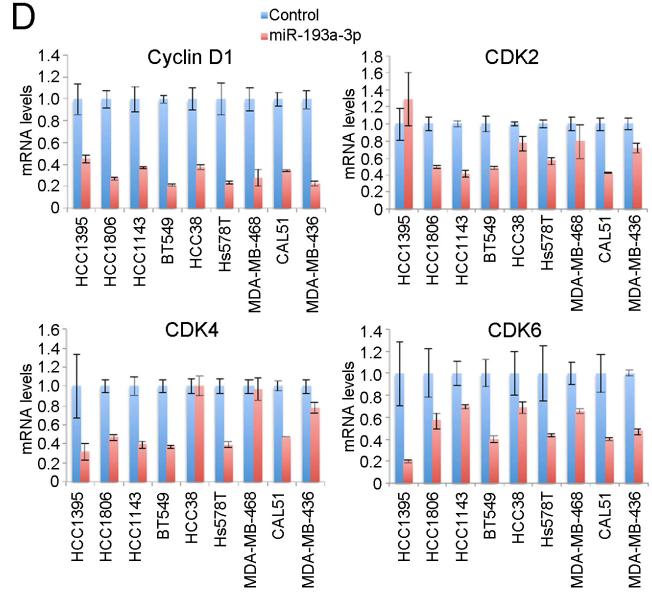
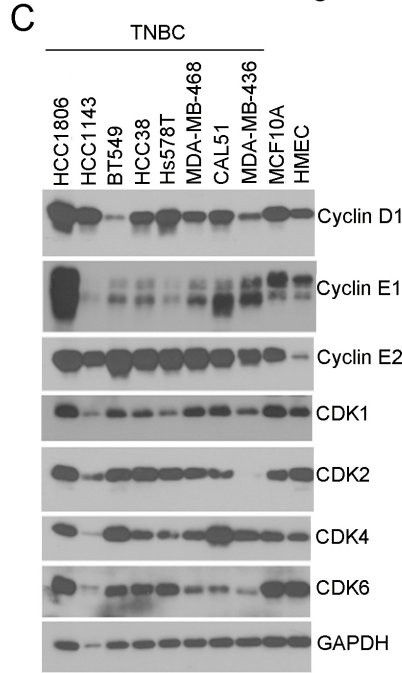
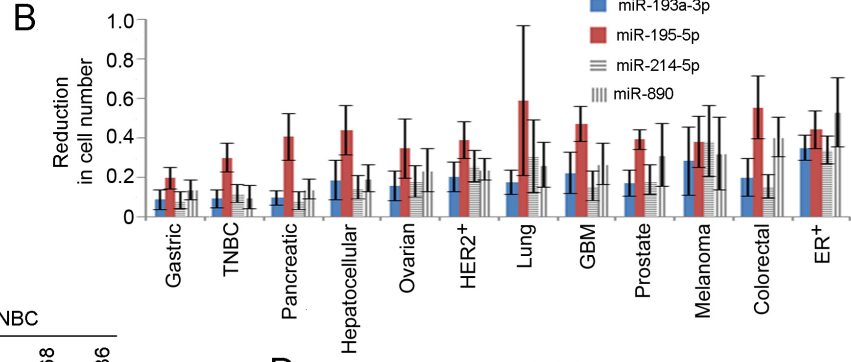
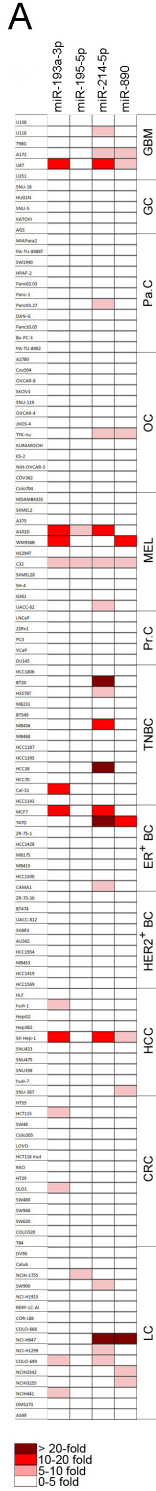


Figure S5, Related to Figures 5 and 6. Response of Cancer Cells to Cell Cycle-Targeting miRNAs

(A) Four cell cycle-targeting miRNAs (miR-193a-3p, miR-195-5p, miR-214-5p, miR-890), or a control miRNA (Ambion, Negative control #2) were ectopically expressed in the indicated human Cancer Cell Line Encyclopedia (CCLE) cell lines, and cells were monitored for senescence (SA- β -galactosidase staining) 6 days post miRNA-transfection. The fold-increase in the number of positive cells is indicated by the red color. Each horizontal row corresponds to the indicated cell line, each vertical column to the indicated cell cycle-targeting miRNA. For each miRNA and cell line, n=3. Cell lines corresponding to the following cancer types were used: GBM, glioblastoma multiforme; GC, gastric cancer; Pa.C, pancreatic cancer; OC, ovarian cancer; MEL, melanoma; Pr.C, prostate cancer; TNBC, triple-negative breast cancer; ER⁺ BC, estrogen receptor-positive breast cancer; HER2⁺ BC, HER2-positive breast cancer; HCC, hepatocellular carcinoma; CRC, colorectal carcinoma; LC, lung carcinoma.

(B) 122 cell lines from CCLE were transfected as in (A) and cell numbers were determined after 6 days. The reduction in cell number was defined as the ratio of cell numbers observed in cells transfected with a given cell cycle-targeting miRNA to cell numbers observed in cells transfected with control miRNA. Results for individual cell lines are shown in Figure 5A and Table S4. In this panel, bars show mean values for cell lines representing the indicated cancer types among 122 cell lines tested, upon expression of miR-193a-3p, miR-195-5p, miR-214-5p, or miR-890. For each miRNA and cell line, n=3. TNBC, triple-negative breast cancer; HER2⁺, HER2-positive breast cancer; GBM, glioblastoma multiforme; ER⁺, estrogen receptor-positive breast cancer.

(C) Immunoblot analysis of cyclins and CDK levels in triple-negative breast cancer cell lines used in Figures 6A and 6B, along with two non-transformed mammary epithelial cell lines (MCF10A and HMEC).

(D) Reduction of the indicated cell cycle targets of miR-193a-3p (cyclin D1, CDK2, CDK4 and CDK6) upon ectopic expression of this miRNA in the indicated triple-negative breast cancer cell lines. Cells were transfected with miR-193a-3p (red bars) or with control miRNA (blue bars), and the levels of the indicated transcripts were measured by RT-qPCR. For each bar, n=3. Please see panel (E) below for reduction of cyclins D3 and E2 levels by miR-193a-3p.

(E) The levels of transcripts for cyclins D1, D3 and E2 in the indicated four triple-negative breast cancer cell lines. For each cell line, parental cells transfected with control miRNA are represented by blue bars and parental cells transfected with miR-193a-3p by red bars. Grey bars, cells engineered by us to ectopically express miRNA-resistant cyclins D1, D3 and E2, transfected with control miRNA (the bars show the sum of endogenous and ectopically expressed cyclins), striped bars, cells ectopically expressing miRNA-resistant cyclins D1, D3 and E2, transfected with miR-193a-3p. For each bar, n=3.

(F) Fold-induction of apoptosis (measured by cleaved caspase 3 assays) in the indicated breast cancer cell lines transfected with control miRNA (blue bars), or with miR-193a-3p (red bars). Grey bars, cells were engineered to express miRNA-resistant cyclins D1, D3 and E2, and transfected with control miRNA or with miR-193a-3p. Shown is fold-induction of apoptosis in miR-193a-3p-transfected cells, relative to cells transfected with control miRNA. For each bar, n=3.

(G) The levels of transcripts for cyclin D1, CDK2 and CDK6 in A549 and LoVo cell lines. For each cell line, blue bars represent parental cells transfected with control miRNA and red bars parental cells transfected with miR-214-5p. Grey bars, cells engineered by us to ectopically express miRNA-resistant cyclin D1, CDK2 and CDK6, transfected with control miRNA (the bars show the sum of endogenous and ectopically expressed cyclin/CDKs), striped bars, cells ectopically expressing miRNA-resistant cyclin D1, CDK2 and CDK6, transfected with miR-214-5p. For each bar, n=3.

In (B) and (D-G) shown are mean values; error bars, SD.

Table S4, Related to Figure 5. Response of 122 Human Cancer Cell Lines from The Cancer Cell Line Encyclopedia to Four Cell Cycle-Targeting miRNAs. Provided as an Excel file.

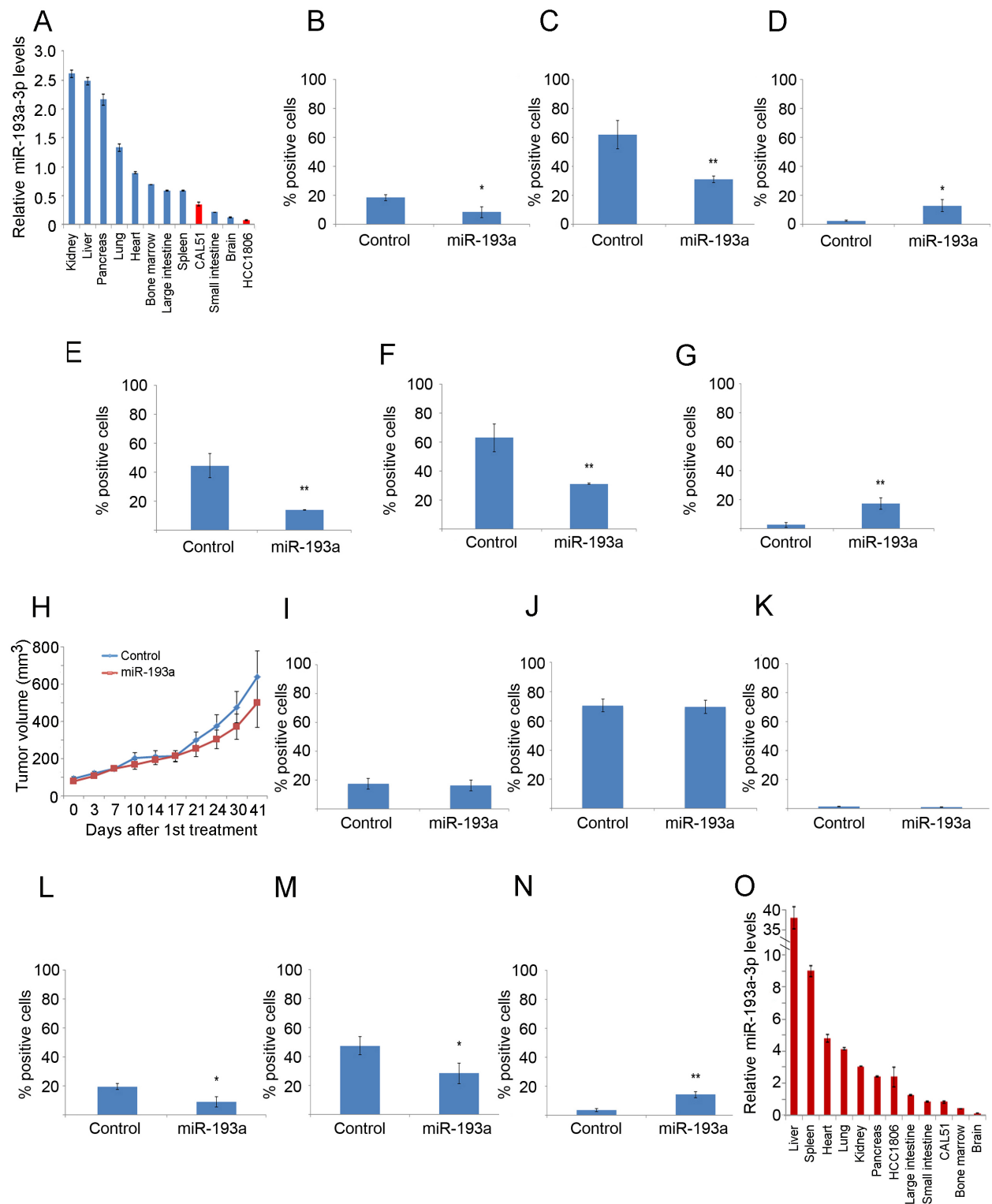


Figure S6, Related to Figure 7. In Vivo Administration of miR-193a-3p Results in Reduced Tumor Cell Proliferation and Increased Apoptosis of Triple Negative Breast Cancers

(A) Expression levels of the endogenous miR-193a-3p in the indicated normal mouse organs (blue bars) and in tumors derived from CAL51 and HCC1806 cells (red bars), relative to the ubiquitously expressed and highly abundant small RNA U6. N = at least 3 mice per tumor/organ.

(B-D) Quantification of BrdU-positive (B), Ki67-positive (C), and cleaved caspase 3-positive cells (D) in CAL51 breast cancer xenografts, from mice treated with nanoparticles containing control miRNA, or containing miR-193a-3p. Each bar represents quantification of tumors from three mice.

(E-G) Similar analyses as in (B-D) of HCC1806 breast cancer cell xenografts.

(H) Mice bearing xenografts derived from CAL51 cells engineered by us to ectopically express miRNA-resistant cyclins D1, D3 and E2 were systemically treated with nanoparticles containing control miRNA or miR-193a-3p. Shown are mean tumor volumes at the indicated days after starting the treatment, error bars, SEM. N=10 per group (control and miR-193a-3p, experiment repeated 2 times).

(I-K) Quantification of BrdU-positive (I), Ki67-positive (J), and cleaved caspase 3-positive cells (K) in CAL51 breast cancer xenografts expressing miRNA-resistant cyclins, from mice treated with nanoparticles containing control miRNA or containing miR-193a-3p (from panel H). Each bar represents quantification of tumors from five mice.

(L-N) Similar analyses as in panels (B-D) and (E-G) using xenografts of patient-derived triple-negative breast cancer. Staining for BrdU (L), Ki67 (M), or cleaved caspase 3 (N).

(O) Analysis of the uptake of miR-193a-3p into the indicated mouse organs and into xenografted tumors derived from CAL51 and HCC1806 cells, relative to the ubiquitously expressed and highly abundant small RNA U6. Bars represent miR-193a-3p levels in miR-193a-3p-treated mice. N = at least 3 mice per tumor/organ.

In (A-G) and (I-O) shown are mean values; error bars, SD. Statistical test through unpaired t test *, $p < 0.05$; **, $p < 0.01$.

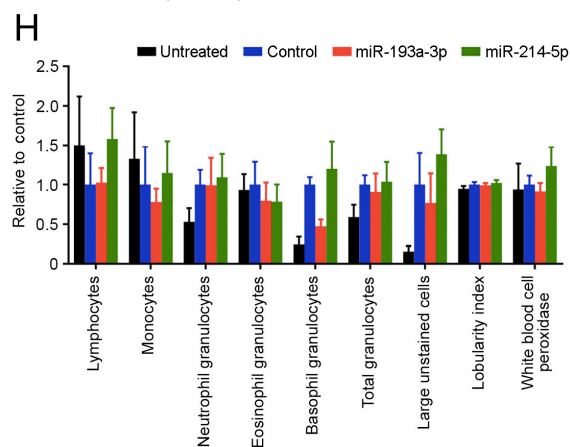
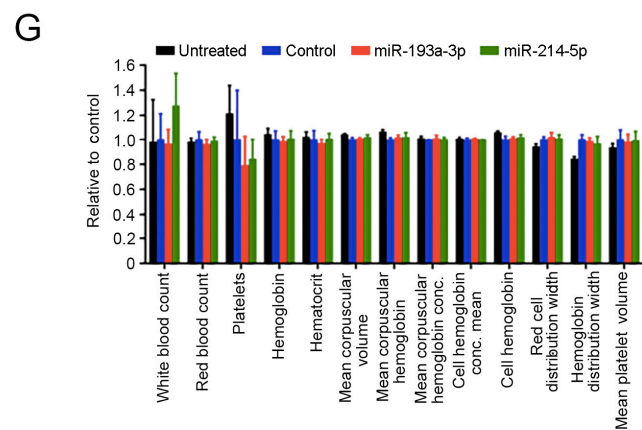
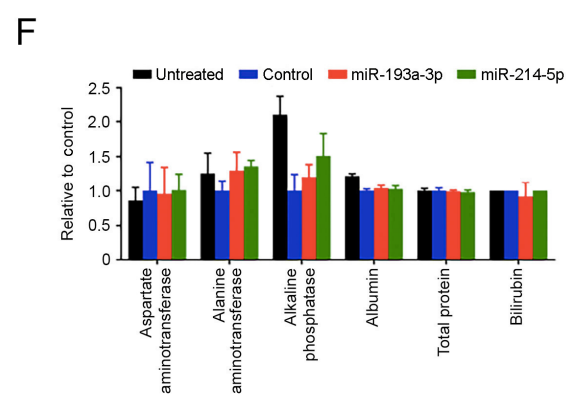
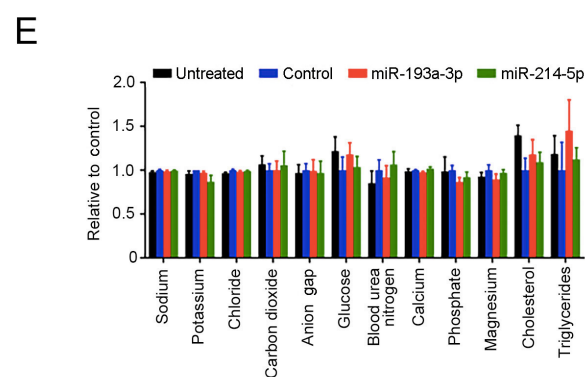
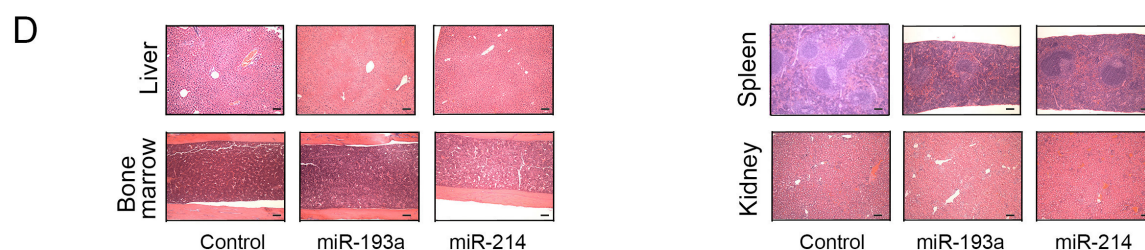
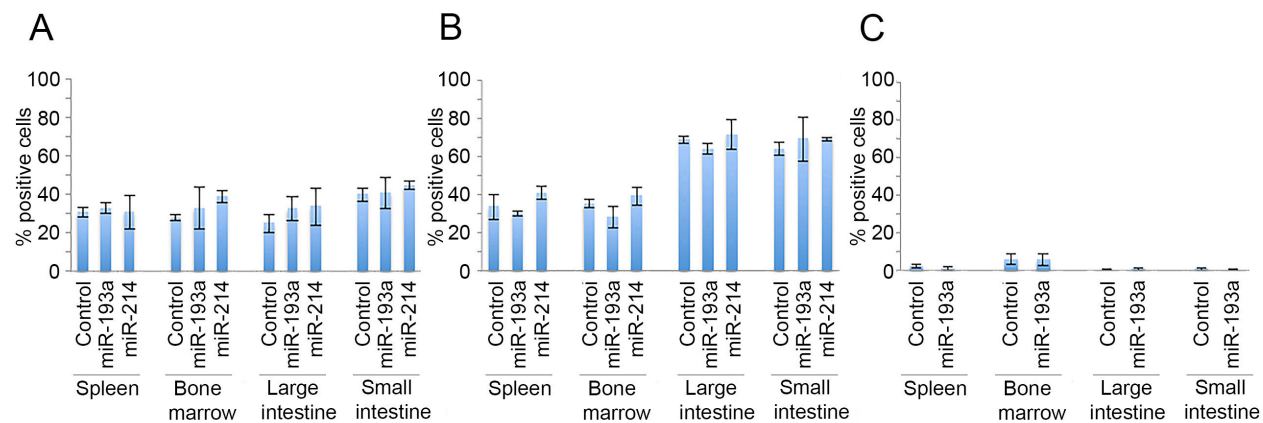


Figure S7, Related to Figures 7 and 8. No Obvious Effects on Normal Organs Upon In Vivo Administration of miR-193a-3p or miR-214-5p

(A-C) Quantification of BrdU-positive (A), Ki67-positive (B), and cleaved caspase 3-positive cells (C) in the indicated organs of mice treated for 23 days with nanoparticles containing control miRNA (Ambion, Negative control #1), or miR-193a-3p, or miR-214-5p. Each bar represents quantification of organs from at least three mice.

(D) Examples of hematoxylin and eosin-stained sections from the indicated organs of mice treated for 23 days with nanoparticles containing control miRNA, or miR-193a-3p, or miR-214-5p. N=3 mice per group. Scale bar represents 100 μ m.

(E-H) The indicated parameters were measured in peripheral blood of untreated mice, or mice treated for 17 days with nanoparticles containing control miRNA, or miR-193a-3p, or miR-214-5p. N=6 mice per group.

In (A-C) and (E-H) shown are mean values; error bars, SD.

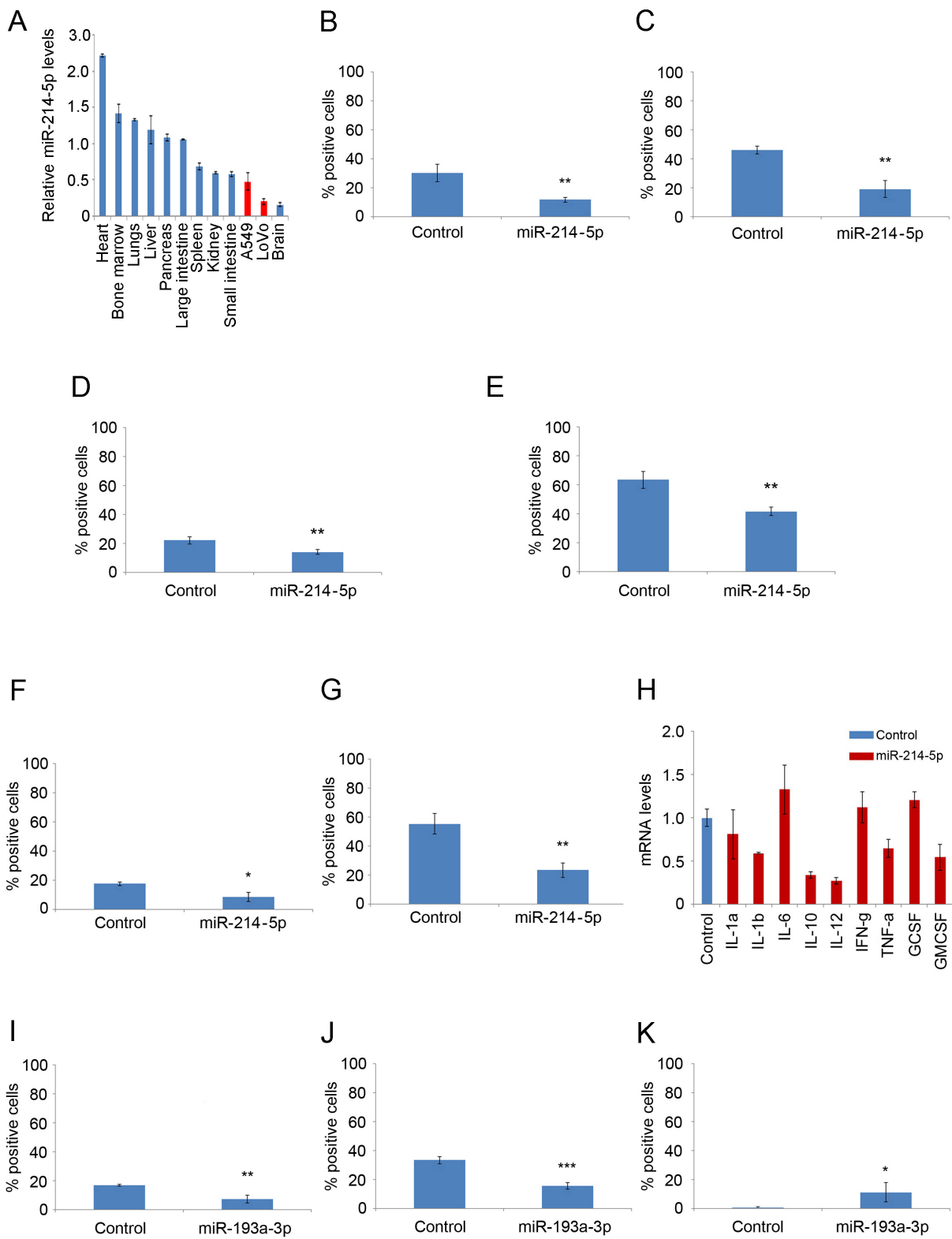


Figure S8, Related to Figure 8. In Vivo Administration of miR-214-5p or miR-193a-3p Results in Reduced Tumor Cell Proliferation of *KRAS*-Mutant Cancers and Sarcoma

(A) Expression levels of the endogenous miR-214-5p in the indicated normal mouse organs (blue bars) and in tumors derived from A549 and LoVo cells (red bars), relative to the ubiquitously expressed and highly abundant small RNA U6. N = at least 3 mice per tumor/organ.

(B and C) Quantification of BrdU-positive (B), and Ki67-positive (C) cells in *KRAS*-mutant A549 lung cancer cell xenografts, from mice treated with nanoparticles containing control miRNA (Ambion, Negative control #1), or containing miR-214-5p. Each bar represents quantification of tumors from three mice.

(D and E) Similar analyses as in (B and C) of LoVo *KRAS*-mutant colorectal cell xenografts.

(F and G) Similar analyses as above using xenografts of patient-derived *KRAS*-mutant colorectal adenocarcinoma.

(H) Analysis of mRNA levels of the indicated nine cytokines in livers of mice treated with nanoparticles containing control miRNA (Ambion, Negative control #1), or containing miR-214-5p, two days post-injection, Each bar represents quantification from three mice. IL-1a, Interleukin 1 α ; IL-1b, Interleukin 1 β ; IL-6, Interleukin 6; IL-10, Interleukin 10; IL-12, Interleukin 12; IFN- γ , Interferon γ ; TNF- α , Tumor necrosis factor α ; GCSF, granulocyte-colony stimulating factor; GMCSF, granulocyte-macrophage colony-stimulating factor. mRNA levels were measured using the delta-delta-Ct method with GAPDH as the reference.

(I-K) Quantification of BrdU-positive (I), Ki67-positive (J), and cleaved caspase 3-positive cells (K) in xenografts of patient-derived dermatofibrosarcoma protuberans, from mice treated with nanoparticles containing control miRNA, or containing miR-193a-3p. Each bar represents quantification of tumors from three mice.

In (A-K) shown are mean values; error bars, SD. Statistical test through unpaired t test. *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$.

Table S5, Related to Figure 8. Expression-Based Algorithm to Predict Response of Primary Tumors to Cell Cycle-Targeting miRNAs. Provided as an Excel file.

Table S6, Related to STAR methods. Provided as an Excel file.

Table S7, Related to STAR methods. Provided as an Excel file.

weight. Wound healing was also unaffected by treatment. The importance of target expression levels was addressed in an elegant experiment investigating the threshold of cell-surface CD276 molecules required for PBD-ADC to exert its cytotoxic effects; this was found to be above 20,000 antibody-binding sites (Seaman et al., 2017). These results suggest that expression level may be a key aspect to consider when targeting such tumor or tumor endothelial markers, which will determine the balance of specificity against toxicity and, hence, the therapeutic index. The results of Seaman et al. are also encouraging for the prospect of targeting other tumor or tumor vasculature markers where there are concerns of widespread expression.

Finally, this study identifies the use of staining antibodies that provide a starting

point for stratification of patients into groups likely to respond to these ADC drugs based on relative expression of CD276, or lack thereof. Provided phase I and II trials do not indicate unacceptable toxicity, the results of Seaman et al. could open a new era of ADC therapeutics, with ramifications for targeting the tumor vasculature beyond anti-angiogenic inhibition of the VEGF pathway.

REFERENCES

- Bridgeman, V.L., Vermeulen, P.B., Foo, S., Bilecz, A., Daley, F., Kostaras, E., Nathan, M.R., Wan, E., Frentzas, S., Schweiger, T., et al. (2017). *J. Pathol.* 241, 362–374.
- Burrows, F.J., and Thorpe, P.E. (1993). *Proc. Natl. Acad. Sci. USA* 90, 8996–9000.
- Diamantis, N., and Banerji, U. (2016). *Br. J. Cancer* 114, 362–367.
- Donnem, T., Hu, J., Ferguson, M., Adighibie, O., Snell, C., Harris, A.L., Gatter, K.C., and Pezzella, F. (2013). *Cancer Med.* 2, 427–436.
- Elinzano, H., Hebda, N., Luppe, D., Turchetti, W., Rosati, K., Sikov, W.M., and Safran, H. (2016). *J. Clin. Oncol.* 34.
- Frentzas, S., Simoneau, E., Bridgeman, V.L., Vermeulen, P.B., Foo, S., Kostaras, E., Nathan, M.R., Wotherspoon, A., Gao, Z.H., Shi, Y., et al. (2016). *Nat. Med.* 22, 1294–1302.
- Müller, P., Kreuzaler, M., Khan, T., Thommen, D.S., Martin, K., Glatz, K., Savic, S., Harbeck, N., Nitz, U., Gluz, O., et al. (2015). *Sci. Transl. Med.* 7, 315ra188.
- Picarda, E., Ohaegbulam, K.C., and Zang, X. (2016). *Clin. Cancer Res.* 22, 3425–3431.
- Seaman, S., Stevens, J., Yang, M.Y., Logsdon, D., Graff-Cherry, C., and St Croix, B. (2007). *Cancer Cell* 11, 539–554.
- Seaman, S., Zhu, Z., Saha, S., Zhang, X.M., Yang, M.Y., Hilton, M.B., Morris, K., Szot, C., Morris, H., Swing, D., et al. (2017). *Cancer Cell* 31, this issue, 501–515.

Not So Fast: Cultivating miRs as Kinks in the Chain of the Cell Cycle

Matthew J. Schiewer^{1,5} and Karen E. Knudsen^{1,2,3,4,5,*}

¹Department of Cancer Biology

²Department of Medical Oncology

³Department of Urology

⁴Department of Radiation Oncology

⁵The Sidney Kimmel Cancer Center

Thomas Jefferson University, Philadelphia, PA 19107, USA

*Correspondence: karen.knudsen@jefferson.edu

<http://dx.doi.org/10.1016/j.ccell.2017.03.012>

In this issue of *Cancer Cell*, Hydbring and colleagues define a novel class of microRNAs (miRNAs), deemed “cell-cycle-targeting miRNAs,” that target several cyclins/CDKs, reduce tumor cell growth, and induce apoptosis. These miRNAs effectively suppressed chemoresistant patient-derived xenograft growth in vivo, and efficacy could be prospectively predicted with an expression-based algorithm.

One of the hallmarks of cancer is unbridled cell proliferation, often manifesting through alterations of the cell-cycle machinery, including cyclin-dependent kinases (CDKs) and their cognate catalytic partners, the cyclins (Hanahan and Weinberg, 2011). Cyclin/CDK complexes regulate cell-cycle phase transitions through selective phosphorylation of substrates that control all aspects of cell division, including commitment to completion of a

mitotic cell cycle, achieved through phosphorylation and inactivation of the retinoblastoma (RB) tumor suppressor protein. These functions are counterbalanced by canonical endogenous CDK inhibitors and tumor suppressors. Loss of these cell-cycle-attenuating tumor suppressors, or, by contrast, deregulation of cyclins, CDKs, or CDK targets, can afford a mechanism for tumor cells to bypass the appropriate cell-cycle regulation and

occur at high frequency in human malignancies (Otto and Sicinski, 2017). As such, developing effective therapeutic means to thwart cell-cycle dysregulation may lead to significant benefit for cancer patients.

The study by Hydbring et al. brings understanding to the role of miRNAs in controlling cell-cycle regulatory proteins and supports the concept that miRNA function could be harnessed as a means

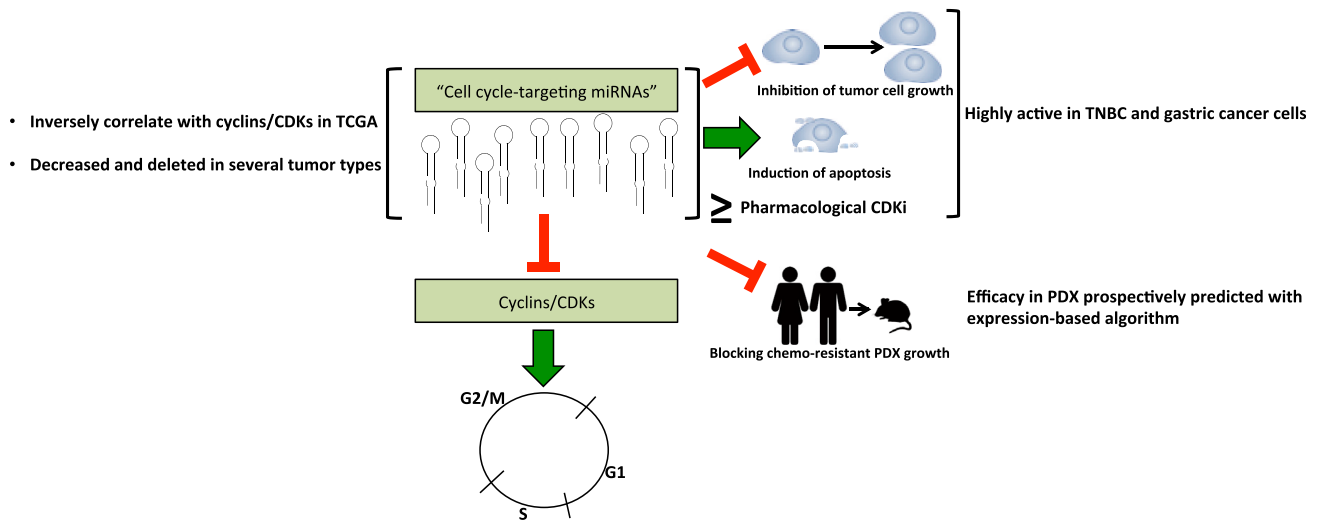


Figure 1. Summary of Hydbring et al.

Through several screens, it was determined that a subset of miRNAs target cyclins and CDKs, which are positive drivers of the cell cycle. These “cell-cycle-targeting miRNAs” were found to be inversely correlated with cyclins/CDKs in The Cancer Genome Atlas (TCGA) data, as well as either reduced or deleted in several tumor types. Therapeutic efficacy of these cell-cycle-targeting miRNAs was enriched in TNBC and gastric cancer cell lines, which was validated in chemotherapy-resistant PDX models.

to counteract unchecked cell-cycle progression (summarized in Figure 1) (Hydbring et al., 2017). miRNAs are short, regulatory RNA species that reduce the expression of gene products by inducing mRNA degradation or inhibition of translation (Lagos-Quintana et al., 2001). Similar to the cell-cycle machinery, miRNAs show frequent alteration in human malignancy (Lu et al., 2005). Understanding of miRNA function is a fertile ground of current investigation, and a number of potential applications in both diagnostics and therapeutics have been realized (Adams et al., 2017; Hydbring and Badalian-Very, 2013). Hydbring et al. screened for miRNAs that target nine cyclins and CDKs of importance for cell division in cancer cells. The results of this screen revealed 30 miRNAs with the capacity to target five or more cyclins/CDKs and 14 miRNAs capable of targeting six or more cyclins/CDKs, thus demonstrating a novel class of miRNAs that regulate nearly all components of the cell-cycle machinery. The number of miRNAs that target multiple cyclins/CDKs is significantly higher than predicted by chance alone. Furthermore, prioritization studies identified anti-correlation of selected miRNAs with anticipated cyclin/CDKs in human tumors, and the miRNAs were found to be downregulated or deleted in several tumor types.

Building upon these initial findings, a second screen utilizing a 122-cancer-cell-line library comprised of 12 tumor types was employed to test the efficacy of these cell-cycle-targeting miRNAs. Four miRNAs identified in the initial screen demonstrated significant growth-inhibitory effects in the cell lines examined, albeit to differing magnitudes. Strikingly, cells derived from gastric cancers and triple-negative breast cancers (TNBCs) demonstrated a more robust response, potentially due to greater reliance on the targeted cyclins/CDKs. In these tissue types, it was further noted that cell-cycle-targeting miRNAs harbored the capacity to induce apoptosis, suggesting that the miRNAs elicit not only the expected cytostatic effects but also cytotoxic responses. Investigation of the underlying basis for enhanced responsiveness in TNBC and gastric cancer may therefore lay the foundation for determining which tumors may be most responsive to miR-based therapies targeting cell-cycle regulation.

Recent clinical successes targeting CDK activity have increased enthusiasm for refining the means to suppress cell-cycle control in a subset of tumor types, most particularly as related to CDK4/6 inhibitors (CDK4/6i). The CDK4/6i palbociclib is FDA approved for the treatment of estrogen receptor (ER)-positive, HER2-negative advanced breast cancer in

combination with estrogen deprivation or ER-targeted therapies (letrozole or fulvestrant, respectively) (Finn et al., 2015; Turner et al., 2015), based in part on clinical trial data demonstrating that the combination resulted in improved progression-free survival (PFS) compared to hormone therapy alone. More recently, the CDK4/6i ribociclib was FDA approved for this same patient population in combination with letrozole, based on clinical data indicating that the combination lengthened PFS (Hortobagyi et al., 2016). In principle, the studies here have the potential to build on the success of CDK4/6i, as the miRNAs identified proved capable of simultaneously suppressing more than five to six CDKs or cyclins and therefore could provide a more potent mechanism to suppress the cancer cell cycle. Furthermore, whereas CDK4/6i are approved for ER-positive disease, the data from Hydbring et al. identified ER-negative tumor models (TNBC) as exceptional responders to miRNA-based cell-cycle suppression. Head-to-head comparison revealed that miR193a-3p is more effective than palbociclib in blocking TNBC cell growth, and miR193a-3p elicits anti-tumor effects even in the context of RB loss, which renders cells resistant to palbociclib. miR193a-3p was also more effective than lapatinib in HER2-positive cells and was comparable to the combination of

lapatinib and palbociclib. While these findings require validation and development of translational potential, the study by Hydbring et al. provides robust evidence that cell-cycle-targeting miRNAs hold promise as anti-cancer agents that may hold advantages over CDK4/6 targeting alone in breast cancer.

Harnessing the miRNAs for translational potential yielded promising results in patient-derived xenograft (PDX) models of chemotherapy-resistant TNBC and colorectal cancer. In both systems, miRNA-containing nanoparticles were effective in eliciting anti-tumor responses, whereas normal tissues were spared any toxicity. In a doxorubicin- and taxane-resistant TNBC PDX, nanoparticles containing miR193a-3p reduced proliferation and induced apoptosis, demonstrating that cell-cycle-targeting miRNA therapy may be effective in chemotherapy-resistant tumors. Furthermore, in a KRAS mutant FOLFOX- and FOLFIRI-resistant colorectal tumor PDX, miR214-5p blocked tumor growth while sparing normal tissues and was more effective in KRAS mutant than wild-type cells. These data indicate that tumor phenotype and/or genotype may influence the utility of cell-cycle-targeting miRNAs. Given the demonstrated selectivity, it is intriguing to speculate that the heightened reliance of tumor cells on cyclins/CDKs can be effectively targeted in malignant tissues using the cell-cycle-targeting miRNA strategy.

Hydbring et al. also developed an expression-based algorithm to predict responses to cell-cycle-targeting miRNAs.

As proof of principle, the authors transcriptionally profiled an aggressive dermatofibrosarcoma protuberans that was resistant to both Gleevec and PI3K inhibition. Utilizing the developed algorithm, it was predicted that the tumor would be responsive to miR-193a-3p. Subsequent administration of nanoparticles containing miR193a-3p to mice harboring a PDX of the dermatofibrosarcoma protuberans resulted in significant reduction of tumor growth, demonstrating the utility of the algorithm. Based on these exciting findings, future studies examining combinatorial strategies with cell-cycle-targeting miRNAs may lead to effective means of managing malignancy.

There are currently three clinical trials utilizing miRNA-based therapies listed on www.clinicaltrials.gov. Two (NCT01829971, NCT02862145) were recently terminated due to immune-related serious adverse events, suggesting that a proper nomination schema, as might be done with the expression-based algorithm described by the authors, may be needed to optimize miRNA-based therapies. The third therapeutic miRNA trial is a phase I trial utilizing a miR-16 mimic in progressive mesothelioma and non-small-cell lung cancer (NCT02369198), which is employing an epidermal growth factor receptor-targeting mechanism. A targeting strategy may be needed to circumvent the possible toxicity of miRNA-based cancer therapy, although the work by Hydbring et al. demonstrates that use of nanoparticles to deliver cell-cycle-targeting miRNAs does not elicit toxicity in immunocompromised mice. Importantly, while several studies

that have examined the capacity of miRNA-based technology to be clinically useful report mixed results, the strategy defined by the authors provides hope for using cell-cycle-targeting miRNAs for cancer management in the near future.

REFERENCES

- Adams, B.D., Parsons, C., Walker, L., Zhang, W.C., and Slack, F.J. (2017). *J. Clin. Invest.* 127, 761–771.
- Finn, R.S., Crown, J.P., Lang, I., Boer, K., Bondarenko, I.M., Kulyk, S.O., Ettl, J., Patel, R., Pinter, T., Schmidt, M., et al. (2015). *Lancet Oncol.* 16, 25–35.
- Hanahan, D., and Weinberg, R.A. (2011). *Cell* 144, 646–674.
- Hortobagyi, G.N., Stemmer, S.M., Burris, H.A., Yap, Y.S., Sonke, G.S., Paluch-Shimon, S., Campone, M., Blackwell, K.L., André, F., Winer, E.P., et al. (2016). *N. Engl. J. Med.* 375, 1738–1748.
- Hydbring, P., and Badalian-Very, G. (2013). *F1000Res.* 2, 136.
- Hydbring, P., Wang, Y., Fassl, A., Li, X., Matia, V., Otto, T., Choi, Y.J., Sweeney, K.E., Suski, J.M., Yin, H., et al. (2017). *Cancer Cell* 31, this issue, 576–590.
- Lagos-Quintana, M., Rauhut, R., Lendeckel, W., and Tuschl, T. (2001). *Science* 294, 853–858.
- Lu, J., Getz, G., Miska, E.A., Alvarez-Saavedra, E., Lamb, J., Peck, D., Sweet-Cordero, A., Ebert, B.L., Mak, R.H., Ferrando, A.A., et al. (2005). *Nature* 435, 834–838.
- Otto, T., and Sicinski, P. (2017). *Nat. Rev. Cancer* 17, 93–115.
- Turner, N.C., Ro, J., André, F., Loi, S., Verma, S., Iwata, H., Harbeck, N., Loibl, S., Huang Bartlett, C., Zhang, K., et al.; PALOMA3 Study Group (2015). *N. Engl. J. Med.* 373, 209–219.